Coulomb's Law & Electric Field

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Coulomb's Law & Electric Field

Electrostatics deals with the study of electric charges at rest. Matter comprises of certain elementary particles. Most common three elementary particles are protons, neutrons and electrons.

The electromagnetic force between the charged particles is one of the fundamental forces in nature. In electrostatics, we will study the forces, fields and potentials arising due to electric charges at rest.

Electric Charge

Electric charge is a fundamental property of matter which produces and experiences electric and magnetic effects.

Units of charge

SI unit of charge is "coulomb (C)".

Besides the SI units, there are some other units for charge also.

Two other units are "electrostatic unit (esu)" and "Faraday".

where, 1 C = 3×10^9 esu.

1 Faraday = 96500 C.

Also, 1 C is a very large amount of charge, so some smaller units are also used sometimes.

1 micro-coulomb (μC) = 10^{-6} C.

1 nano-coulomb (nC) = 10^{-9} C.

1 pico-coulomb (pC) = 10^{-12} C.

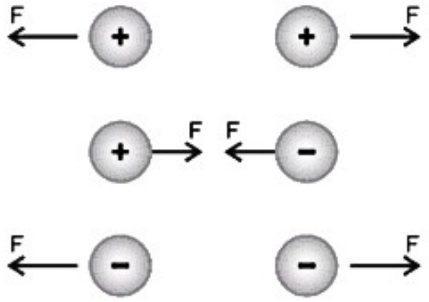
The smallest possible value of transferable charge is the amount of charge on an electron (e = 1.6×10^{-19} C). Charges occur on a body when electrons are transferred from or to it. Note that electrons are in outer shells of an atom and are loosely bound. So, their transfer can occur by various methods explained in upcoming sections. In no case, transfer of protons can take place as protons are inside the nucleus of an atom.

Properties of electric charges

(1) Charge is a scalar quantity.

(2) Charges are of two types; one is positive and other is negative (given by Benjamin Franklin).

(3) Like charges repel each other and opposite charges attract each other.



Points to remember

The total charge is conserved but total number of charge particles is not conserved.

A quark is a type of elementary particle and fundamental constituent of matter. They have charges of ±e/3 or ±2e/3. These particles are always found to be in combination.

(4) Charge is additive. Charges are always added algebrically with signs.

If a system contains two point charges q_1 and q_2, then

Total charge on the system = $(q_1 + q_2)$

(5) Charge is quantized.

Charge exists in discrete units equal to the integral multiple of electronic charge (charge on an electron).

If Q is the total charge on a body, then $Q = \pm (ne)$

Where, "n" is a non negative integer and "e" is the charge on 1 electron.

For any amount of charge (Q) on a body, the value of "n" must be a non negative integer.

Consider the figure shown below. There are two neutral spheres A and B. Now, if one electron from A is transferred to B, then A becomes positively charged by losing an electron while B becomes negatively charged by gaining an electron.

Charge on A, $Q_A = +e$

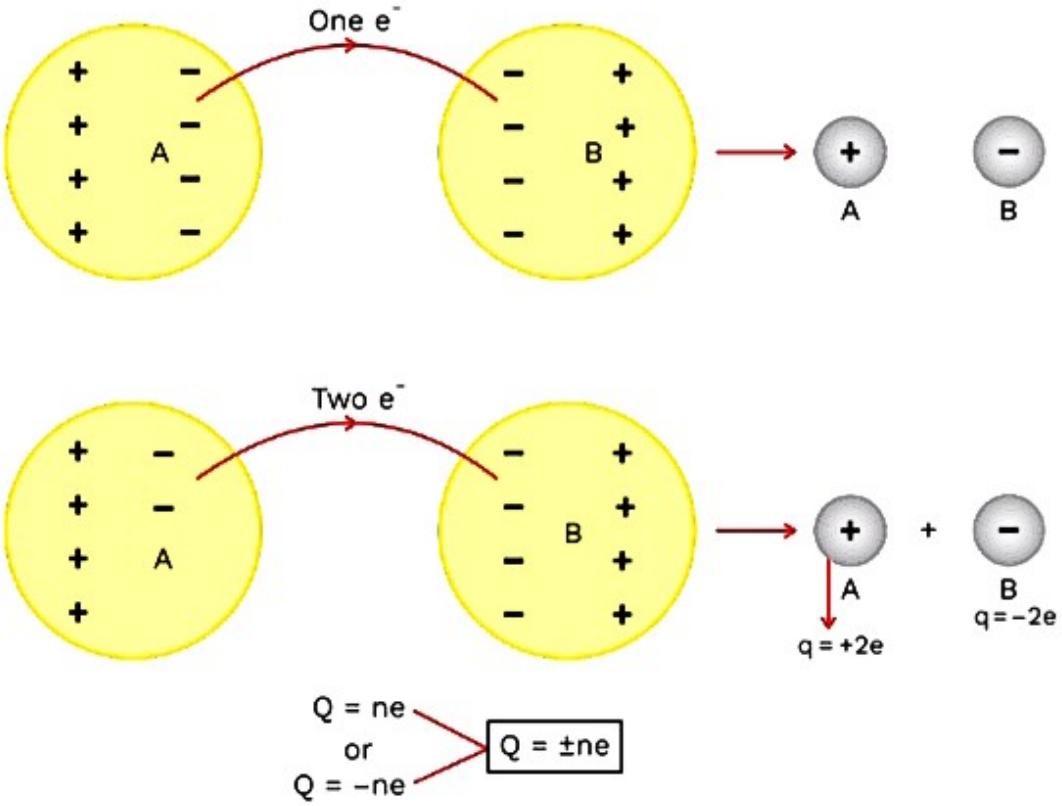
Charge on B, $Q_B = -e$

Similarly, if two electrons are transferred from A to B, then

Charge on A, $Q_A = +2e$

Charge on B, $Q_B = -2e$

So, we can have a general formula for transfer of n electrons as, $Q = \pm ne$.



(6) Charge is conserved.

The total charge of an isolated system is always conserved. It is possible to create or

destroy charged particles inside an isolated system, but it is not possible to create or destroy 'net charge' on the system.

For example,

"In a beta decay process, a neutron converts itself into a proton and a fresh electron is created. The charge, however, remains zero before and after the event".

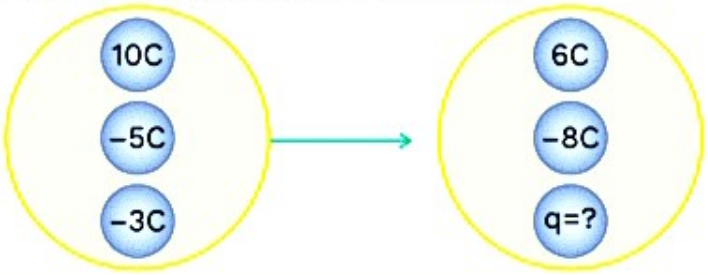
$$ {}_0n^1 \xrightarrow[\text{Process}]{\beta\text{-decay}} p^+ + {}_{-1}\beta^0 + \bar{\nu} $$

or $-1e^0$

Anti neutrino

Initial charge	Final charge
+ve = 0	+ve = $+1e^-$
−ve = 0	−ve = $-1e^-$
net = 0	net = 0

Q. **Find the value of q on the system if the system is isolated and the charge re-arrangement takes place as shown in the figure.**



Sol. As the system is isolated, so net charge on the system remains conserved,

$10 - 5 - 3 = 6 + (-8) + q$

$2 = -2 + q$

$q = 4C$

Q. **Mark-out the correct option:**

(a) The total charge of the universe is constant.

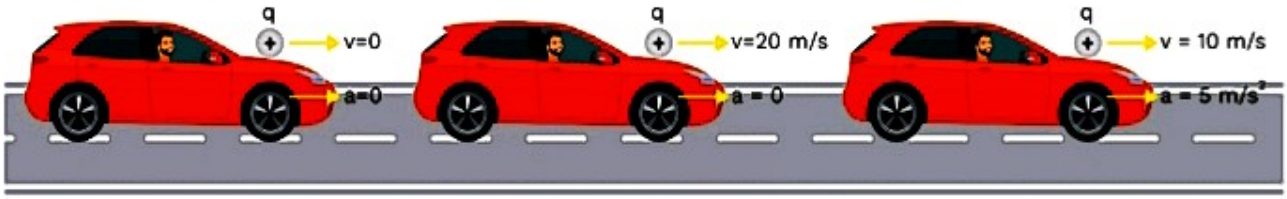
(b) The total positive charge of the universe is constant.

(c) Total negative charge of the universe is constant.

(d) The total number of charged particles in the universe is constant.

Sol: (a) is the correct answer as universe is an isolated system and net charge in an isolated system remains constant.

(7) Charge is invariant in nature.
Charge does not depend on reference frame. It means that the value of charge is independent of frame of reference.



Same charge is observed by all observers in each case

Note:

Mass and length are variant in nature.
When the speed of an object becomes comparable to the speed of light, the mass and length of a body vary as given by the formula in the figure below.

m_0 (rest mass)



$$m = \frac{m_0}{\sqrt{1-\frac{v^2}{c^2}}}$$

$$L = L_0\sqrt{1-\frac{v^2}{c^2}}$$

Points to remember

- According to Einstein mass variation formula $m = \frac{m_0}{\sqrt{1-\frac{v^2}{c^2}}}$.
  But no such variation in charge is observed.
- When a neutrino (an elementary charged particle) and an antineutrino collide, they annihilate (destroy) each other and produce two gamma photons.

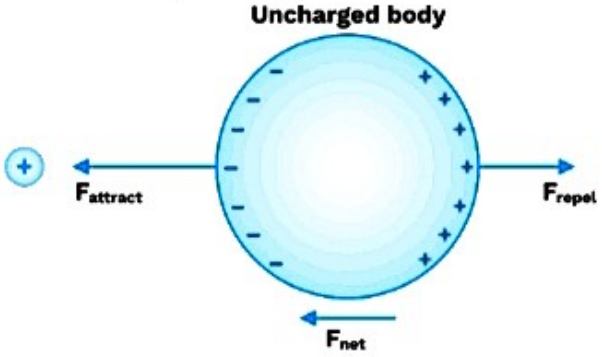
(8) Mass can exist without charge (like neutron) but charge can't exist without mass.

(9) Charge produces electric field, magnetic field and radiations.

	Electric Field	Magnetic Field	Radiations
Stationary Charge	✓	✗	✗
Charge moving with constant velocity	✓	✓	✗
Charge moving with variable velocity	✓	✓	✓

(10) A charged particle/body can attract an uncharged body.

On placing a charged particle near an uncharged conducting body, it polarises the charges on the surface of the uncharged body. The net charge is still zero but the negative charges on the body are closer to the charged particle as compared to the positive charges on the body. As the distance between positive and negative charges is less, the attractive force is greater than the repulsive force and hence there is attraction between the charged particle and the uncharged body.



(11) A charged particle/body can't attract an uncharged particle. (Polarisation is not possible in a particle)

Note:

Two particle/bodies can attract due to many reasons, but repulsion only exists if both have charges. Therefore, "repulsion" is true test of "electrification".

(12) Generally, when a body becomes negatively charged, its mass increases.

(In general, a body is charged by transferring of electrons. Negatively charged bodies have some excess number of electrons which slightly increase the mass of the body.)

(13) Generally, when a body becomes positively charged, its mass decreases (due to deficiency of electrons).

Q. **Which of the following values of charge can be possessed by a body?**
(A) 3.2×10^{-19} C (B) -0.8×10^{-18} C (C) 2.4×10^{-19} C (D) 1×10^{-19} C

Sol: (A) $n = \frac{Q}{e} = \frac{3.2 \times 10^{-19}}{1.6 \times 10^{-19}} = 2$ (As n is an integer, so it is possible.)

(B) $n = \frac{Q}{e} = \frac{-0.8 \times 10^{-18}}{1.6 \times 10^{-19}} = -5$ (As n is an integer, so it is possible.)

(C) $n = \frac{Q}{e} = \frac{2.4 \times 10^{-19}}{1.6 \times 10^{-19}} = 1.5$ (As n is not an integer, so it is not possible.)

(D) $1 \times 10^{-19} < 1.6 \times 10^{-19}$ (It is not possible.)

Q. **Find the charge constituted by all the electrons in 8 grams of helium.**

Sol: Molar mass of He = 4 g/mol

No. of moles of He in 8 gm = $\frac{8}{4} = 2$

No. of He atoms in 8 gm = $(2 \times 6.023 \times 10^{23})$
1 He atom contains 2 electrons.

So, total number of electrons = $(2 \times 2 \times 6.023 \times 10^{23}) = 24.092 \times 10^{23}$
Charge constituted by all the electrons

$$= e \times \left(24.092 \times 10^{23}\right) = \left(1.6 \times 10^{-19}\right) \times \left(24.092 \times 10^{23}\right) = 3.85 \times 10^{5}\ C$$

Some fundamental charged particles	
Positive Charged Particles	**Negative Charged Particles**
1. Proton (p^+)	**1. Electron (e^-, β^-)**
Charge = $+1.6 \times 10^{-19}$ C	Mass = 9.1×10^{-31} kg
Mass = 1.672×10^{-27} kg	Charge = 1.6×10^{-19} C
Composition = 2 up quarks (u), 1 down quark (d)	Composition = Elementary particle
2. Alpha (α^{++}) particle	**2. Neutron (n^0)**
Mass = 4 amu	Charge = 0 e
Charge = $2 \times (1.6 \times 10^{-19}$ C)	Mass = 1.674×10^{-27} kg
	Note ($m_n > m_p$)
	Composition = 1 up quark, 2 down quarks.
3. Positron (e^+ or β^+)	**3. Photon**
The antimatter counter part of the electron	Charge = 0
Mass = 9.1×10^{-31} kg	Mass = 0
Charge = $+1.6 \times 10^{-19}$ C	Composition = Elementary
Note Up quark = $\frac{2}{3}e$, Down quark = $-\frac{1}{3}e$	

Conductor

Materials having large number of free electrons (electrons which can flow freely inside the material) are called conductors.

Example: All metals like silver, gold, copper, etc are conductors.

Infact, liquids and gases can also show conductivity due to presence of ions like Na^+, Cl^-, H^+, etc.

Insulator

Materials in which free charge carriers are absent are called insulators.

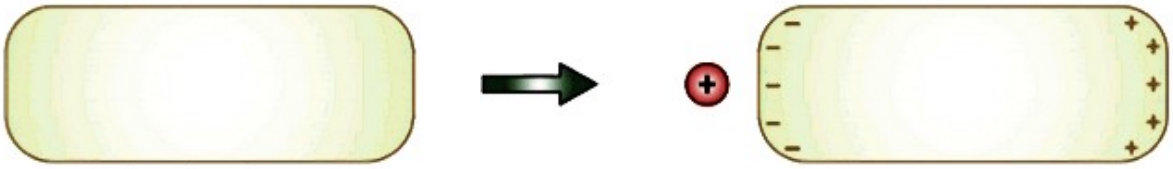
Example: Wood, plastic, etc.

Methods of charging

(a) Polarisation

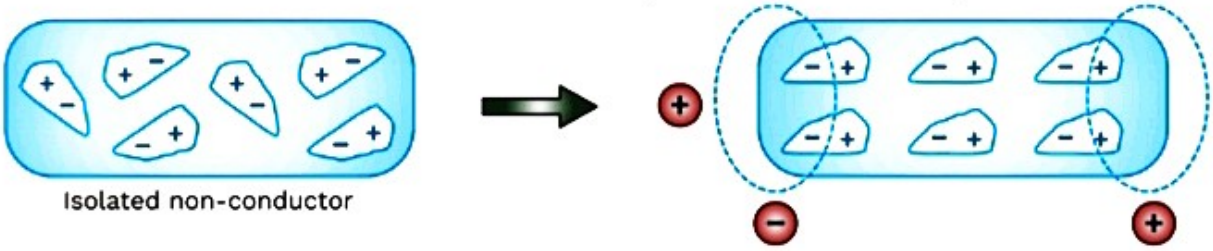
The separation of positively and negatively charged particles in an object due to influence of an electric field or another charged particle/body is called polarisation of charges.

Polarisation of conductor due to presence of external charge



Isolated conductor

Polarisation of non-conductor due to presence of external charge



Isolated non-conductor

(b) Charging by Friction

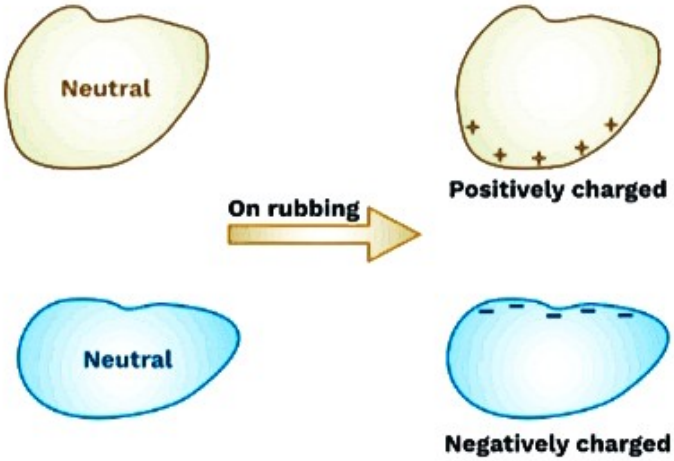
If we rub one body with another body, electrons are transferred from one body to the other.

Both the bodies get different charges in the process.

Note:

More electronegative material gains electrons. On gaining electrons, mass increases slightly.

Less electronegative material loses electrons. On losing electrons, mass decreases slightly.



Two different bodies can be charged on rubbing (friction). If both bodies are made of same material, then they cannot be charged on rubbing.

Also, we cannot charge two conductors (same or different) by rubbing them.

For example, when an **ebonite** (very hard rubber) rod is rubbed against a **fur cloth**, the ebonite rod becomes positively charged and the fur becomes negatively charged.

Positive Charge	Negative Charge
Glass rod	Silk cloth
Woollen cloth	Rubber, Amber, Plastic
Dry hair	Comb
Cat skin	Ebonite rod

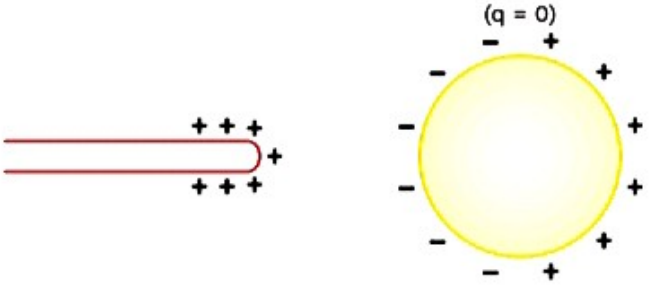
(c) Charging by Induction:

If a charged body is brought near a neutral body, the charged body will attract opposite charges and repel similar charges present in the neutral body.

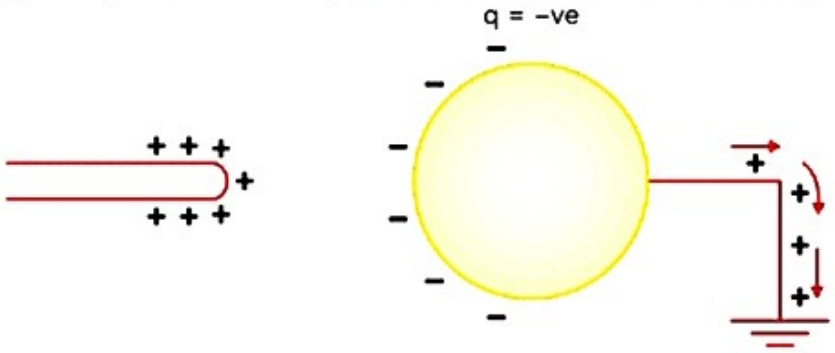
Induction is the phenomenon of redistribution of charges on an uncharged body (by induction) when any other charged body is brought near it.

Charging a body by induction

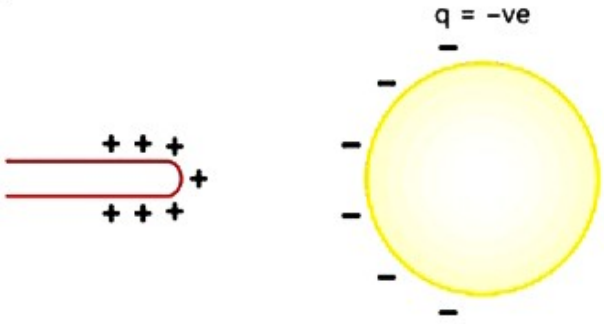
Step-1: Charged body is brought near uncharged conducting body. As a result, the positive and negative charges on the body get separated, but the net charge remains zero.



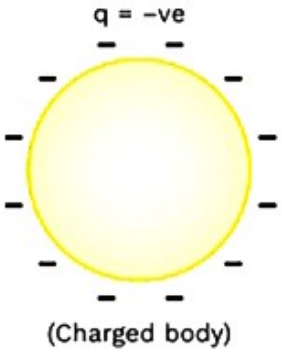
Step-2: Now connect the uncharged body to earth. The positive charges on the other side are neutralised due to connection with earth. Earth is assumed to be a source of infinite charge. So, electrons flow from Earth to neutralise the positive charge on the other side. In general, it is said that positive charge flowed from body to earth.



Step-3:
Disconnect the body from earth.



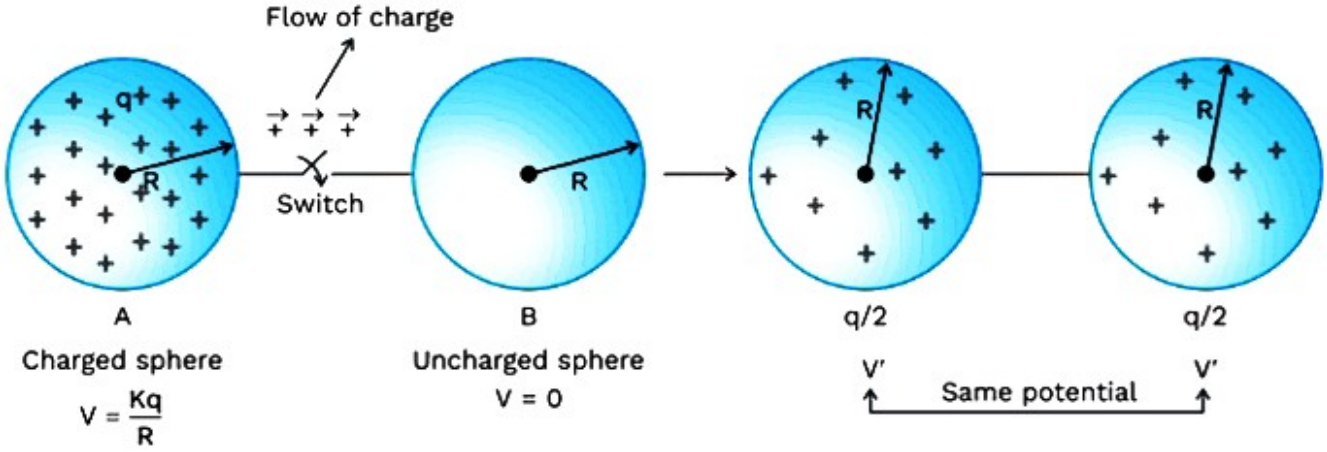
Step-4:
Now remove the charged rod and then the negative charges get distributed on the body. The body gets negatively charged now.



(Charged body)

(d) Charging by Conduction

When two conductors are connected directly, then charges flow from one to the other until potentials of both the conductors become equal. (Potential is explained later in chapter)

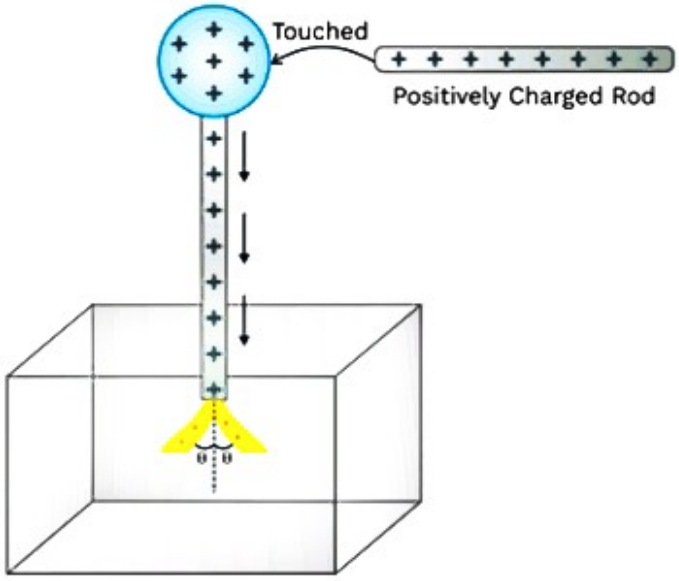


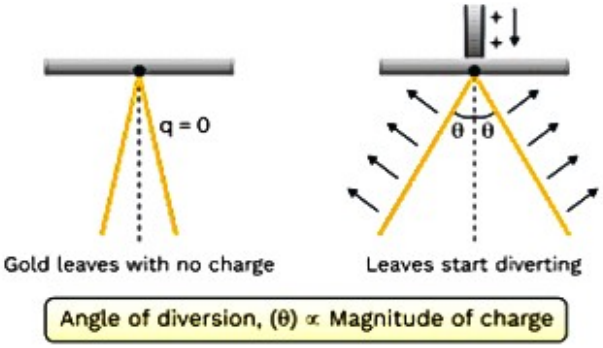
Gold leaf electroscope:

1. Gold leaf electroscope is a simple apparatus to detect charge on a body.

2. It consists of a vertical metal rod housed in a box with thin gold leaves attached to its bottom end. The body whose charge is to be determined is touched to the sphere attached at the top of the electroscope. As the charge flows to the vertical rod in the apparatus, the gold leaves divert, thereby indicating the charge.

   Angle of diversion between the leaves ∝ Magnitude of charge on the body.





Angle of diversion, (θ) ∝ Magnitude of charge

Coulomb's Law

"Charles Augustin de coulomb" a French physicist, began his career as a military engineer. Coulomb gave quantitative law about the force of attraction or repulsion between two point charges. He used a torsional balance (a sensitive device to measure force) for measuring the force between charges.

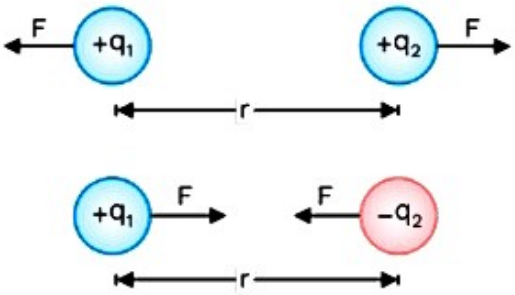
Statement of Coulomb's Law

Coulomb measured the force between two point charges and found that it varied inversely as the square of the distance between the charges and was directly proportional to the product of magnitude of the two charges.
According to Coulomb's law,

$F \propto \frac{1}{r^2}$ ... (inverse square law)

$F \propto |q_1| . |q_2|$

$$F = \frac{K.|q_1|.|q_2|}{r^2} = \frac{1}{4\pi \epsilon_0} \frac{|q_1|.|q_2|}{r^2}$$



where, q_1 = charge on 1st particle.
q_2 = charge on 2rd particle.
K = Coulomb's constant = 9×10^9 Nm^2/C^2

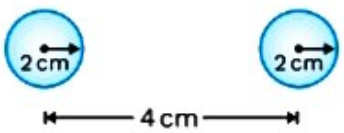
ϵ_0 = Absolute permittivity of vacuum
= 8.85×10^{-12} C^2/Nm^2

Important points to note about Coulomb's law

1. This law is directly valid only for point charges. When the linear size of charged bodies are much smaller than the distance separating them, then the size of the body can be ignored, and the charged bodies are treated as point charges.



We can assume these charges as point charges



We cannot assume these charges as point charges

Things to know

- Coulomb's law is like Newton's law of gravity.
- According to quantum theory, a force varies inversely as the square of the distance if it involves the exchange of particles with no mass.
- Exchange of massless photons is responsible for the electrical force, and exchange of massless gravitons accounts for the gravitational force.

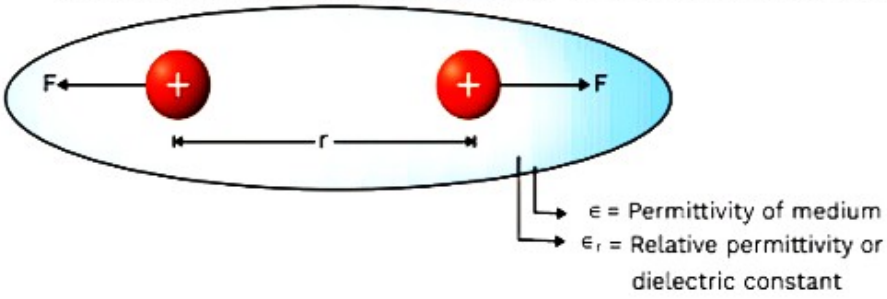
2. It is also known as the 'inverse-square law'.
3. Pair of forces is called 'action reaction' pair, i.e., both forces have same magnitude, and are in opposite direction, acting along the same line, exerted on different particles and appear simultaneously.
4. It holds true for charges which are at rest with respect to each other.
5. Coulomb's force is conservative in nature.

Points to remember

- Coulomb's law is only applicable for point charges.
- The gravitational force between the charged particles is negligible as compared to the electric force between them.
- The force between the two charged particles is a central force.

Electrostatic force between two charges kept in a medium

If two charges are kept in a medium instead of being in vacuum, the permittivity of the medium also comes into play while calculating the force between the charges.



For two charges q_1 and q_2 as shown in the figure,
Net force between the charges,

$$F_{Net} = K \cdot \frac{|q_1| \cdot |q_2|}{r^2}$$

$$F_{Net} = \frac{1}{4\pi\epsilon} \frac{q_1 q_2}{r^2}$$

where, $\epsilon = \epsilon_0 \epsilon_r$

$$\text{So, } F_{net} = \left(\frac{1}{4\pi\epsilon_0\epsilon_r}\right)\frac{q_1 \cdot q_2}{r^2}$$

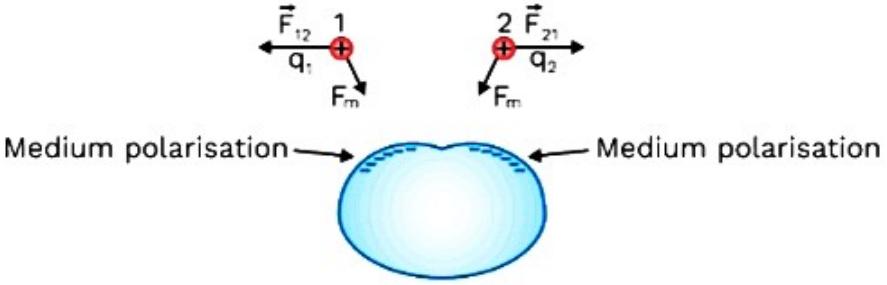
$$= \frac{1}{\epsilon_r} \times \frac{1}{4\pi\epsilon_0} \frac{|q_1 q_2|}{r^2}$$

Points to remember

- Electric force depends on the nature of medium between the charges while gravitational force does not.
- Electric force is conservative. This means work done in a closed loop is zero.
- Net force between the charges in a medium, $\boxed{F_m = \frac{F_0}{\epsilon_r}}$
- Net electric force on both charged particles in presence of medium depends on the presence of medium but force due to one charge on another charge does not depend upon medium.

Force applied by a medium

When a medium is in close vicinity of two point charges, then the medium gets polarised.



Due to polarization, medium also starts applying force on the charges.

$$\vec{F}_{Net} = \vec{F}_0 - \vec{F}_m$$

Net force — Force in vaccum — Force applied by medium

$$\frac{F_0}{\epsilon_r} = F_0 - F_m$$

$$\boxed{F_m = F_0\left(1 - \frac{1}{\epsilon_r}\right)}$$

where, ϵ_r = Relative permittivity of medium or dielectric constant of medium.

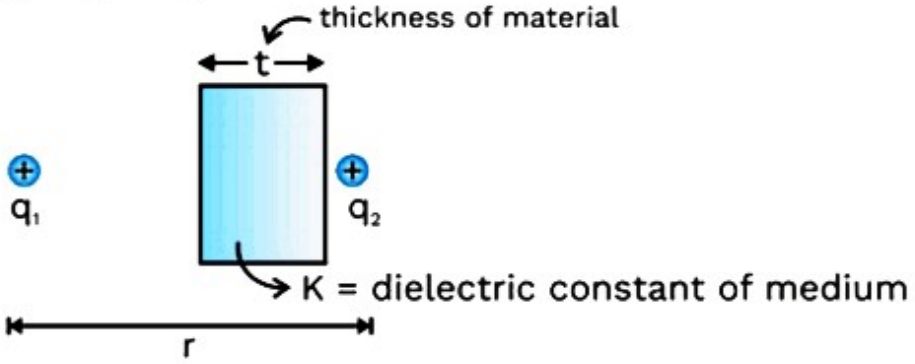
Conclusion

From the above result, it can be seen that the net force (F_{Net}) between the charges decreases. It happens because of the polarization of medium.

Also, note that Coulomb's force between the charges is not affected by any medium. The decrease in the net force is because of an additional force due to medium polarisation.

Force of interaction between charges if space between them is partially filled

Consider the following figure in which a dielectric slab of thickness t is kept in between the two charges q_1 and q_2.



In such a case, the separation (r) between the charges is to be replaced by an effective distance given by, $r' = (r - t) + t\sqrt{k}$

The net force between the charges then becomes,

$$F_{Net} = \frac{1}{4\pi \epsilon_0} \times \frac{q_1 q_2}{r'^2}$$

where, r' = effective distance between the charges.

Physical significance of permittivity

- It represents the capability of a medium to permit electric field.
- It is a physical constant often used in electromagnetism.
- Higher the value of ϵ, lower the value of electric-field due to polarization effect.

Physical significance of permittivity of free space (ϵ_0)

It represents the capability of vacuum to permit electric field.

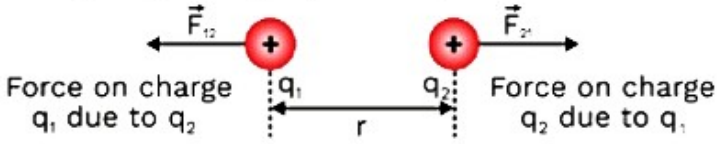
Relative permittivity of medium (ϵ_r)

Relative permittivity of medium is defined as permittivity of medium relative to permittivity of free space.

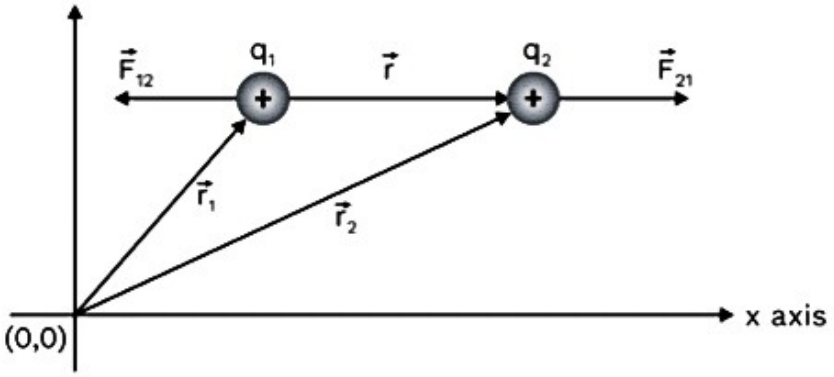
$$\left\{\begin{array}{c} \epsilon_r = 1; \text{ for free space} \\ \epsilon_r = \infty; \text{ for conductors} \\ \epsilon_r > 1; \text{For any other medium} \end{array}\right\}$$

Vector form of Coulomb's Law

Consider two point charges q_1 and q_2 separated by a distance r as shown in the figure.



By drawing a coordinate system as shown below, we can write the forces and position vectors.



Calculation for $\vec{F}_{21}$:

From coulomb's law:

$$F = K\frac{q_1 q_2}{r^2} \text{ (scalar form)}$$

$$\vec{F}_{21} = K\frac{q_1 q_2}{r^2}(\hat{r}_{21}) \qquad \ldots(1)$$

Use triangle law of vector addition

$$\vec{r}_1 + \vec{r} = \vec{r}_2$$

$$\vec{r} = \vec{r}_2 - \vec{r}_1$$

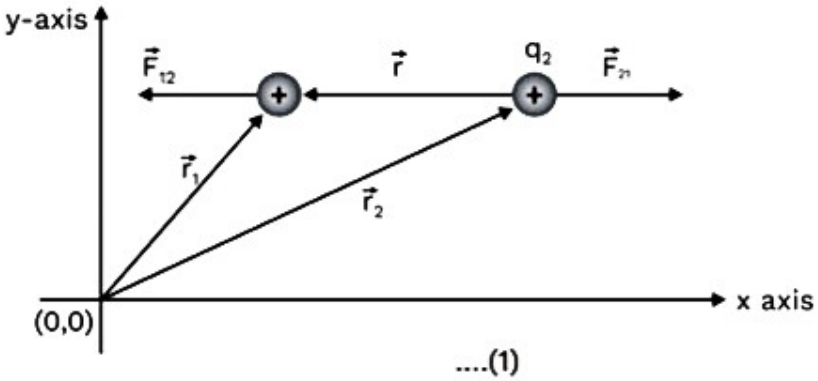
$$|\vec{r}_{21}| = |\vec{r}_2 - \vec{r}_1| = r \text{ (Distance between centre of charges)}$$

$$\hat{r}_{21} = \frac{\vec{r}_{21}}{|\vec{r}_{21}|} = \frac{\vec{r}_2 - \vec{r}_1}{|\vec{r}_2 - \vec{r}_1|} \qquad \ldots(2)$$

From equation (1) and equation (2):

$$\vec{F}_{21} = K\frac{q_1 q_2}{r^2}\frac{(\vec{r}_2 - \vec{r}_1)}{|\vec{r}_2 - \vec{r}_1|}$$

$$\boxed{\vec{F}_{21} = K\frac{q_1q_2\left(\vec{r}_2 - \vec{r}_1\right)}{\left|\vec{r}_2 - \vec{r}_1\right|^3}}$$



Calculation for $\left(\vec{F}_{12}\right)$:

From Coulomb's law:

$$F = K\frac{|q_1q_2|}{r^2}$$

$$\vec{F}_{12} = K\frac{q_1q_2}{r^2}\left(\hat{r}_{12}\right) \quad \text{....(1)}$$

From triangle law of vector addition:

$$\vec{r}_2 + \vec{r} = \vec{r}_1$$

$$\vec{r} = \vec{r}_1 - \vec{r}_2$$

$$\vec{r}_{12} = \vec{r}_1 - \vec{r}_2$$

$$\left|\vec{r}_{12}\right| = \left|\vec{r}_1 - \vec{r}_2\right| = r$$

$$\hat{r}_{12} = \frac{\vec{r}_{12}}{\left|\vec{r}_{12}\right|} = \frac{\vec{r}_1 - \vec{r}_2}{\left|\vec{r}_1 - \vec{r}_2\right|} \quad \text{....(2)}$$

From equation (1) and equation (2)

$$\vec{F}_{12} = K\frac{q_1q_2}{r^2} \times \left(\frac{\vec{r}_1 - \vec{r}_2}{\left|\vec{r}_1 - \vec{r}_2\right|}\right)$$

$$\boxed{\vec{F}_{12} = \frac{Kq_1q_2\left(\vec{r}_1 - \vec{r}_2\right)}{\left|\vec{r}_1 - \vec{r}_2\right|^3}}$$

Also, $\vec{F}_{12} = -K \times \dfrac{q_1q_2\left(\vec{r}_2 - \vec{r}_1\right)}{\left|\vec{r}_1 - \vec{r}_2\right|^3}$

Since, $\left|\vec{r}_1 - \vec{r}_2\right| = \left|\vec{r}_2 - \vec{r}_1\right| = r$

$$\vec{F}_{12} = -\frac{Kq_1q_2\left(\vec{r}_2 - \vec{r}_1\right)}{\left|\vec{r}_2 - \vec{r}_1\right|^3}$$

$$\boxed{\vec{F}_{12} = -\vec{F}_{21}}$$

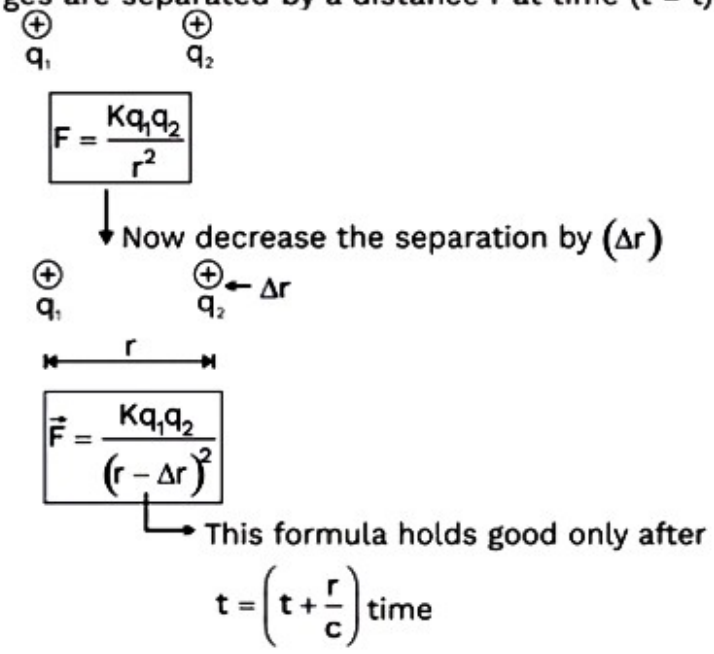
Important Points to note

(1) $\vec{F}_{12} = -\vec{F}_{21}$; These forces are action-reaction pair in accordance with Newton's third law.
(2) Forces are not equal, but the magnitudes of forces are equal.
(3) This force is not affected by presence or absence of any other medium or charge.

Limitations of Coulomb's Law

1. Coulomb's law holds good in the condition when charges are at rest not only at that moment (t = t) but also at the time $t' = \left(t + \frac{r}{c}\right)$

Suppose two charges are separated by a distance r at time (t = t).

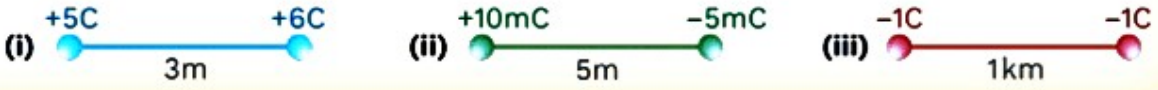


This is so because no information can be transmitted with velocity greater than velocity of light.

2. The forces acting on charges are not always equal and opposite in nature due to propagation speed limit of information.

3. Coulomb's law is valid for point charges only.

Q. **Find the magnitude of electrostatic force acting between the pair of charges shown in the figure below.**



Sol: (i) Force of repulsion, $F = \frac{Kq_1q_2}{r^2}$

$$= \frac{9\times10^9\times5\times6}{3^2} = 3\times10^{10}\,N$$

(ii) Force of attraction, $F = \frac{9\times10^9\times\left(10\times10^{-3}\right)\times\left(5\times10^{-3}\right)}{5^2} = 18000N$

(iii) Force of repulsion, $F = \frac{9\times10^9\times1\times1}{\left(10^3\right)^2} = 9000N$

Q. **The sum of two positive charges is +4 C and are placed at a distance of 1 m from each other. Find the maximum possible repulsive electrostatic force between them.**

Sol: Let one charge has a magnitude x, then the magnitude of the other charge will be (4-x).



$$F = \frac{Kx(4-x)}{1^2} = Kx(4-x)$$

For maximum force, $\frac{dF}{dx} = 0$, and $\frac{d^2F}{dx^2} < 0$.

So, $\frac{dF}{dx} = 4 - 2x = 0 \Rightarrow x = 2$ and $(4 - x) = 2$

Also, $\left(\frac{d^2F}{dx^2}\right)_{x=2} = -2 < 0$

$\therefore F_{max} = \frac{Kx(4-x)}{1} = 9\times10^9\times2\times2 \Rightarrow F_{max} = 36\times10^9\,N$

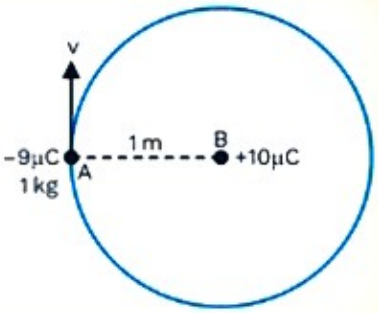
Q. **How much charge should be transferred from A to B such that they repel each other with maximum possible force?**

Sol: From previous solution, we know that force among them will be maximum when both have same charges, i.e., $\frac{(+32)+(2C)}{2} = 17C$ each.



So, (32 – 17) = 15 C charge should be transferred.

Q. **A point charge 'A' is moving in a circle around a fixed charge 'B' as shown. Find the speed 'v' of particle 'A'.**



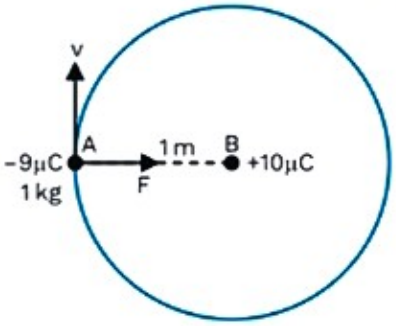
Sol: Required centripetal force = Coulomb force on A

$$m\frac{v^2}{R} = K\frac{q_A q_B}{R^2}$$

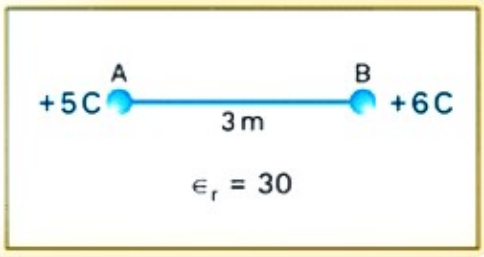
$$v^2 = \frac{9\times10^9\times(9\times10^{-6})(10\times10^{-6})}{1\times1}$$

$$v^2 = 9^2\times10^{-2}$$

$$v = 0.9\ m/s$$



Q. **(i) Find net force on B.**
(ii) Find net force on B due to A.
(iii) Find net force on B due to medium.



Sol: (i) Net force acting on B,

$$F_{net} = \frac{Kq_1q_2}{\varepsilon_r r^2}$$

$$F_{net} = \frac{9\times10^9\times5\times6}{30\times(3)^2} = 10^9 N\ (\rightarrow)$$

(ii) Force acting on B due to A,

$$F_{BA} = \frac{Kq_1q_2}{r^2} = \frac{9\times10^9\times5\times6}{3^2} = 30\times10^9\ \text{N}\ (\rightarrow)$$

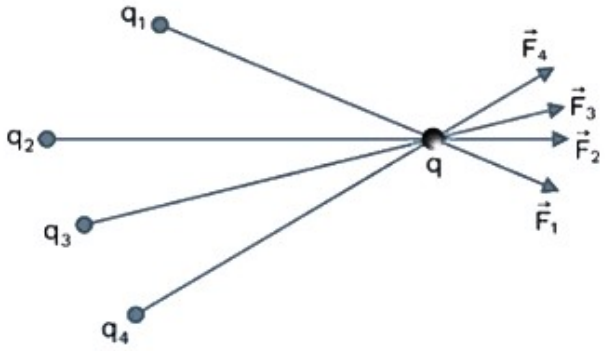
(iii) Force acting on B due to the medium,

$$F_{B,\text{medium}} = F_{BA} - F_{net} = \left(30\times10^9\right) - \left(1\times10^9\right) = 29\times10^9\text{N}\ (\leftarrow)$$

Superposition principle

According to the superposition principle, net force on q will be the vector resultant of all forces, i.e.,

$$\vec{F}_{net} = \vec{F}_q = \vec{F}_1 + \vec{F}_2 + \vec{F}_3 + \vec{F}_4 + \ldots\ldots$$



Q. **Two charges of 5 mC each are placed at (1, 1) m and (4, 5) m, respectively. Find the force acting on both the charges.**

Sol.

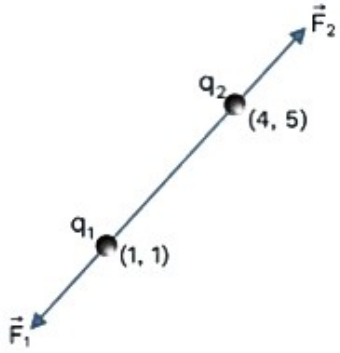
$$\vec{F}_2 = \frac{Kq_1q_2}{\left|\vec{r}_{21}\right|^3}\vec{r}_{21}$$

$$= \frac{\left(9\times10^9\right)\times\left(5\times10^{-3}\right)\times\left(5\times10^{-3}\right)\left[(4-1)\hat{i}+(5-1)\hat{j}\right]}{\left|(4-1)\hat{i}+(5-1)\hat{j}\right|^3}$$

$$= \frac{9\times5\times5\times10^3\left(3\hat{i}+4\hat{j}\right)}{\left(\sqrt{3^2+4^2}\right)^3}$$

$$= \frac{9\times5\times5\times10^3}{5^3}\left(3\hat{i}+4\hat{j}\right)$$

$$\vec{F}_2 = 1800\left(3\hat{i}+4\hat{j}\right)\text{N}$$



Since $\vec{F}_1$ and $\vec{F}_2$ form action-reaction pair,

$$\therefore\ \vec{F}_1 = -\vec{F}_2 \Rightarrow \vec{F}_1 = -1800\left(3\hat{i}+4\hat{j}\right)\text{N}$$

Q. **A charge of 130 mC is placed at (1, 2, 3) m and a charge of −260 mC is placed at (13, 5, 7) m. Find the force acting on the second charge.**

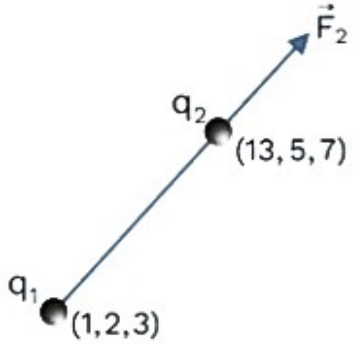
Sol:

$$\vec{F}_2 = \frac{Kq_1q_2}{|\vec{r}_{21}|^3}\vec{r}_{21}$$

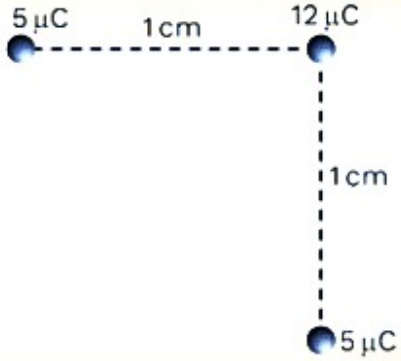
$$= -\frac{9\times10^9\times\left(130\times10^{-3}\right)\times\left(260\times10^{-3}\right)\times\left(12\hat{i}+3\hat{j}+4\hat{k}\right)}{\left|12\hat{i}+3\hat{j}+4\hat{k}\right|^3}$$

$$= -\frac{9\times10^5\times13\times26\times\left(12\hat{i}+3\hat{j}+4\hat{k}\right)}{13^3}$$

$$\vec{F}_2 = -\frac{18}{13}\times10^5\left(12\hat{i}+3\hat{j}+4\hat{k}\right)\text{N}$$



Q. **Find the force acting on 12μC charge.**

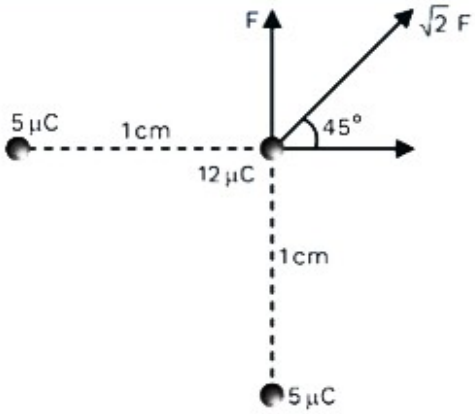


Sol:

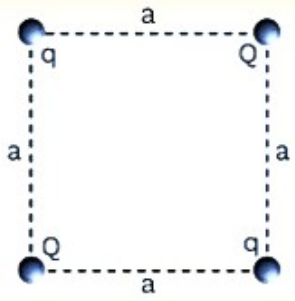
$$F_{net} = \sqrt{2}\,F = \sqrt{2}\,\frac{K(5\,\mu C)(12\,\mu C)}{(1\text{ cm})^2}$$

$$= \frac{\sqrt{2}\times9\times10^9\times60\times10^{-12}}{10^{-4}}$$

$$F_{net} = 54\sqrt{2}\times10^2\text{N}$$



Q. **Find $\left(\frac{Q}{q}\right)$ for which 'Q' will remain in equilibrium.**

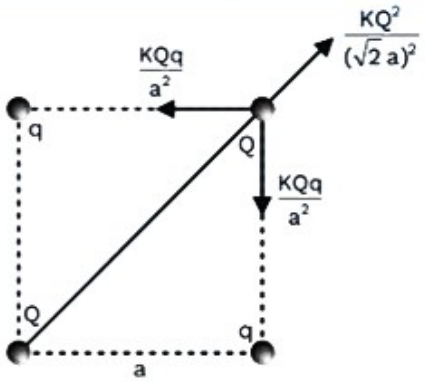


Sol: Both Q and q cannot have the same sign, otherwise all will fly away due to mutual repulsion. Therefore $\left(\frac{Q}{q}\right)$ will be negative.

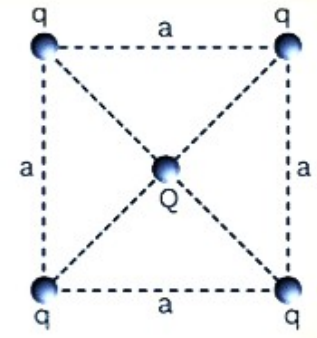
In order to have equilibrium,

$$\frac{KQ^2}{\left(\sqrt{2}a\right)^2} = \sqrt{2}\left|\frac{KQq}{a^2}\right| \quad \Rightarrow \quad \left|\frac{Q}{q}\right| = 2\sqrt{2}$$

$$\Rightarrow \frac{Q}{q} = -2\sqrt{2}$$



Q. **Find $\left(\frac{Q}{q}\right)$ for which q will remain in equilibrium.**

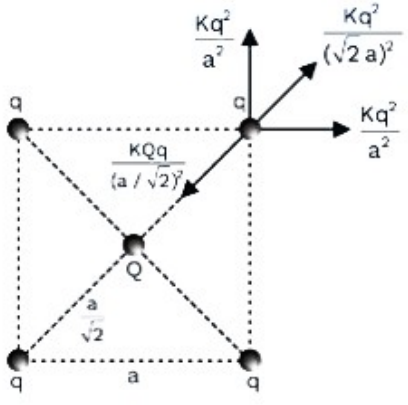


Sol: Both Q and q cannot have the same sign for equilibrium.

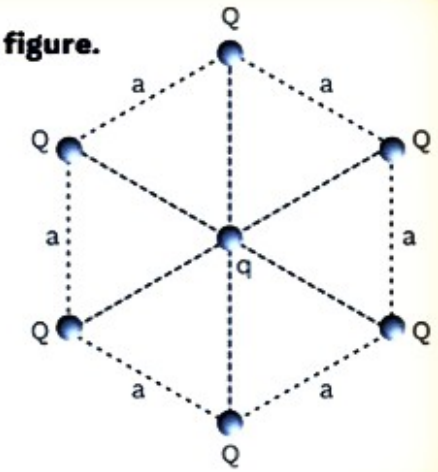
$$\frac{Kq^2}{\left(\sqrt{2}a\right)^2} + \sqrt{2}\frac{Kq^2}{a^2} = \left|\frac{KQq}{\left(a/\sqrt{2}\right)^2}\right|$$

$$\Rightarrow \frac{1}{2} + \sqrt{2} = 2\left|\frac{Q}{q}\right|$$

$$\Rightarrow \left|\frac{Q}{q}\right| = \frac{1}{4} + \frac{1}{\sqrt{2}} \quad \Rightarrow \quad \frac{Q}{q} = -\left(\frac{1}{\sqrt{2}} + \frac{1}{4}\right)$$

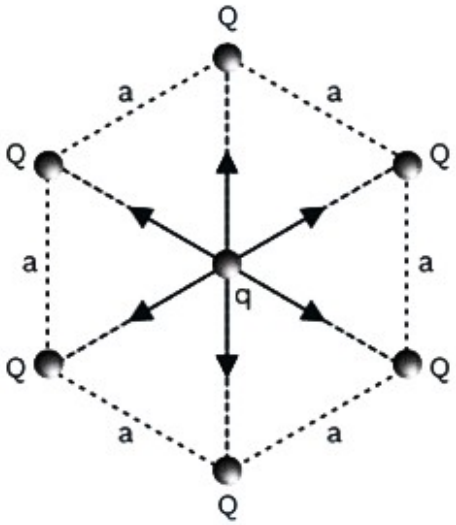


Q. **Find the force acting on charge 'q' in the following figure.**

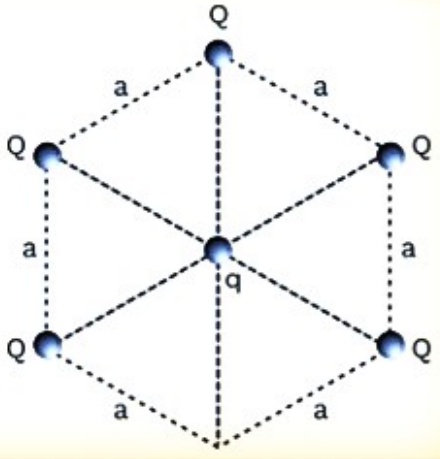


Sol: Due to symmetry, forces acting on the central charge are equal in magnitude and are directed in such a manner that every force cancels out some other force.

Therefore, net force on charge q kept at centre, $F_q = 0$.



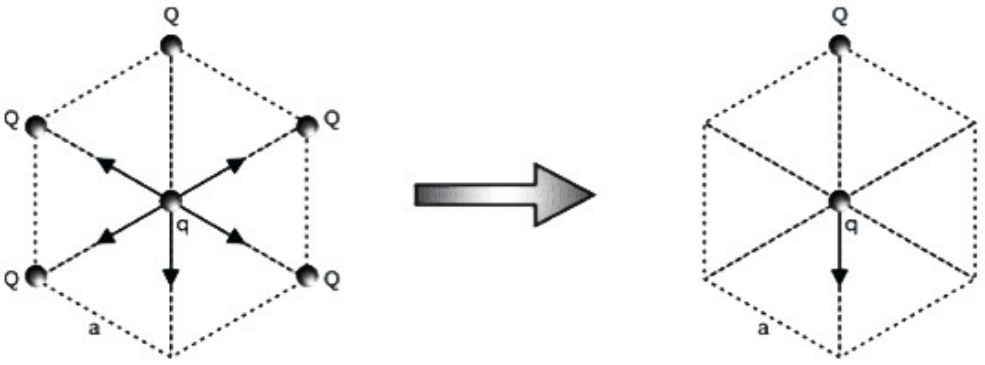
Q. **Find the force acting on charge 'q' in the following figure when one charge is missing from one corner.**



Sol: **Method - 1**

The force due to 5 charges (Q) at each corner is shown acting on the central charge q. If all the 6 corners were occupied by Q, then net charge on q would have been zero. So, the resultant of these five forces would have been balanced by the force due to the missing charge.

Hence, the net force due to 5 charges is equal to the force due to the point charge shown in the second figure.

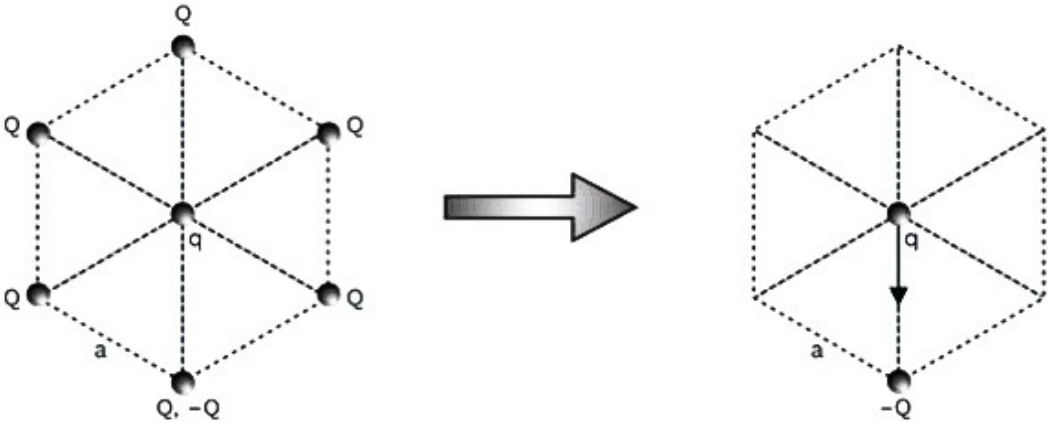


Net force acting on the charge q, $F_q = \frac{KQq}{a^2}$ downwards.

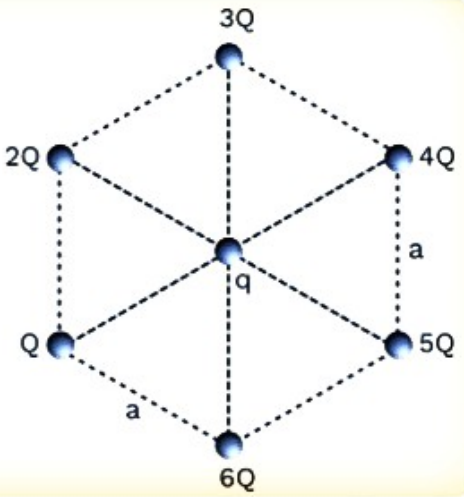
Method - 2

The corner with zero charge can be assumed to be containing +Q and -Q.

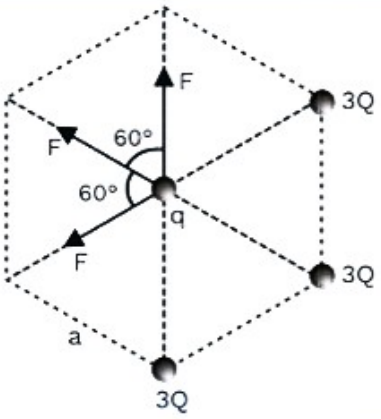
All six +Q charges give a zero net force. So, remaining force on q is due to -Q only which acts downwards.



Q. **Find the force acting on charge 'q' in the following figure.**



Sol: The effective charge and their position w.r.t. force on q, can be shown in figure:

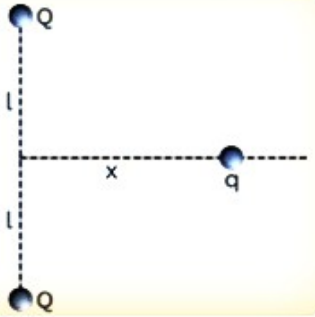


$$F = \frac{K(3Q)q}{a^2}$$

$$F_{net} = F + F\cos 60^\circ + F\cos 60^\circ = 2F$$

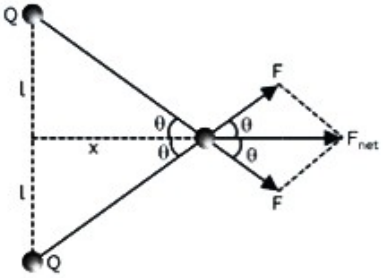
$$F_{net} = 2\left(\frac{K(3Q)q}{a^2}\right) = \frac{6KQq}{a^2}$$

Q. **Find the value of 'x' for which force on 'q' is maximum.**



Sol: Net force on q, $F_{net} = 2F\cos\theta$

$$F_{net} = 2 \times \frac{KQq}{\left(x^2 + \ell^2\right)}\left(\frac{x}{\sqrt{x^2 + \ell^2}}\right) = \frac{2KQqx}{\left(x^2 + \ell^2\right)^{3/2}}$$



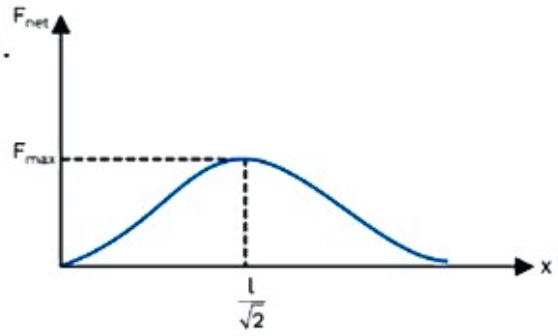
For maximum force, $\frac{dF_{net}}{dx} = 0$ and $\frac{d^2F_{net}}{dx^2} < 0$.

$$\frac{dF_{net}}{dx} = KQq\left[\frac{\left(x^2+\ell^2\right)^{3/2}(1) - x\frac{3}{2}\left(x^2+\ell^2\right)^{1/2}2x}{\left(x^2+\ell^2\right)^3}\right]$$

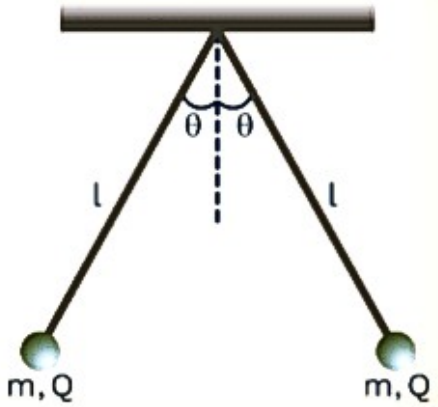
$$= 0$$

$$\Rightarrow x^2 + \ell^2 - 3x^2 = 0$$

$$\Rightarrow x = \pm\frac{\ell}{\sqrt{2}} \Rightarrow x = \frac{\ell}{\sqrt{2}}$$



Q. **Find the value of Q for the given system to be in equilibrium.**



Sol:

Method-1

In equilibrium condition,

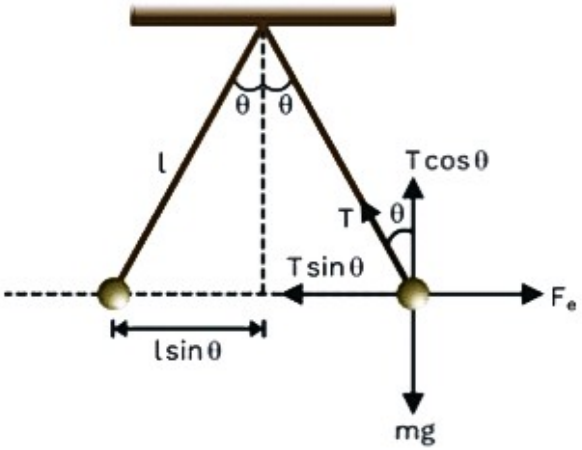
$$T\sin\theta = F_e = \frac{KQ^2}{\left(2\ell\sin\theta\right)^2} \quad ...(1)$$

And,

$T\cos\theta = mg$ ...(2)

$\frac{(1)}{(2)}$ gives $\Rightarrow \tan\theta = \frac{KQ^2}{4\ell^2\sin^2\theta\, mg}$

$$Q = \sqrt{\frac{4\ell^2\sin^2\theta\, mg\tan\theta}{K}}$$



Method-2

In equilibrium,

Along string,

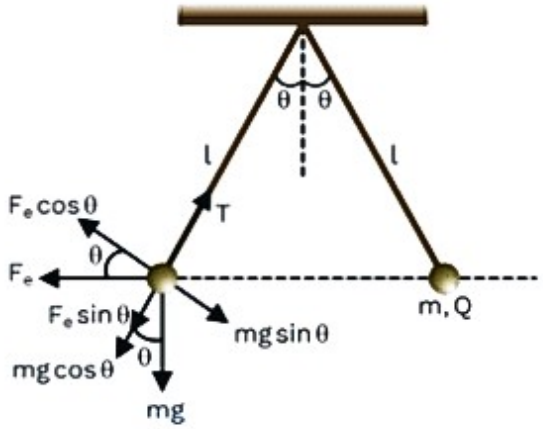
$$T = mg\cos\theta + F_e\sin\theta \quad ...(1)$$

Along tangent,

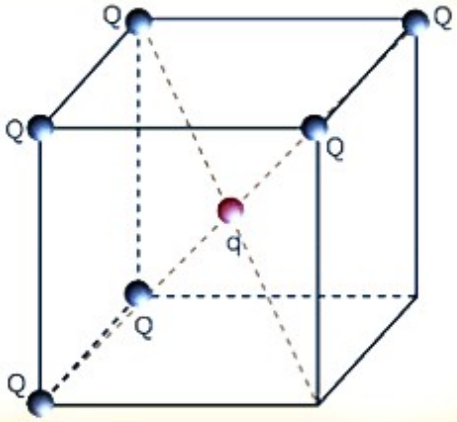
$$mg\sin\theta = F_e\cos\theta$$

$$\Rightarrow mg\sin\theta = \frac{KQ^2}{(2\ell\sin\theta)^2}\cos\theta$$

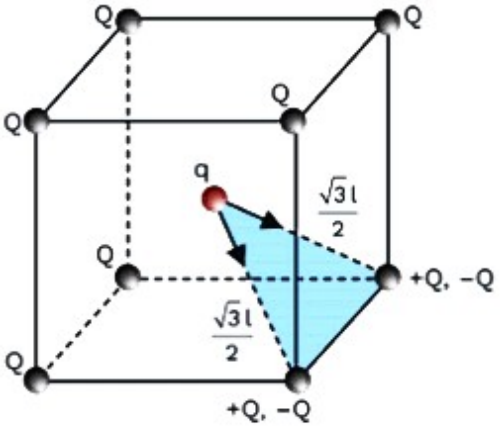
$$\Rightarrow Q = \sqrt{\frac{4\ell^2\sin^2\theta\, mg\tan\theta}{K}}$$

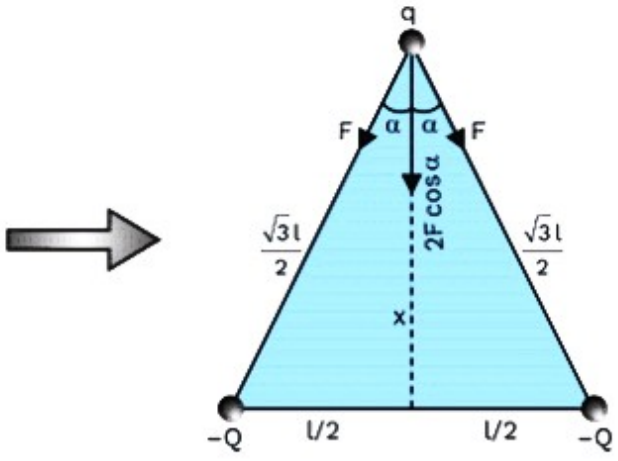


Q. **Find the net force on central charge 'q'.**



Sol: The vacant corners of the cube can be considered to be occupied by +Q, -Q each. Now, 8 corner charges produce net zero force on central charge. So, the net force on the central charge is the resultant force due to -Q and -Q as shown.





From the figure,

$$x^2 + \left(\frac{\ell}{2}\right)^2 = \left(\frac{\sqrt{3}\ell}{2}\right)^2$$

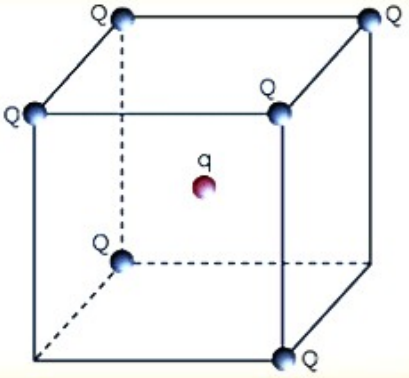
$$x^2 = \frac{3\ell^2}{4} - \frac{\ell^2}{4} = \frac{\ell^2}{2}$$

$$x = \frac{\ell}{\sqrt{2}}$$

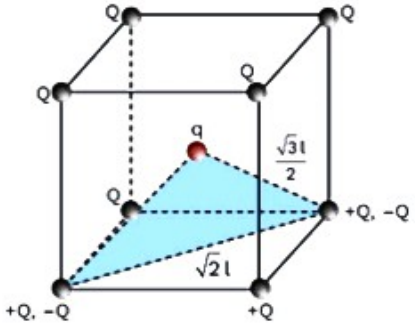
Net force on the central charge, $F_{net} = 2F\cos\alpha = \frac{2KQq}{\left(\frac{\sqrt{3}\ell}{2}\right)^2} \cdot \frac{x}{\left(\frac{\sqrt{3}\ell}{2}\right)}$

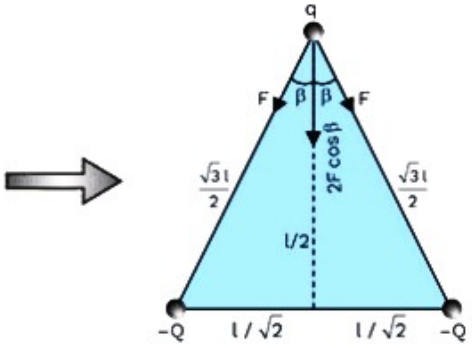
$$F_{net} = \frac{8\sqrt{2}}{3\sqrt{3}} \frac{KQq}{\ell^2}$$

Q. **Find the net force on central charge 'q'.**



Sol: Similar to the previous question,

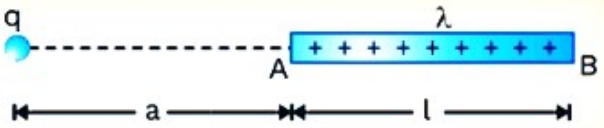




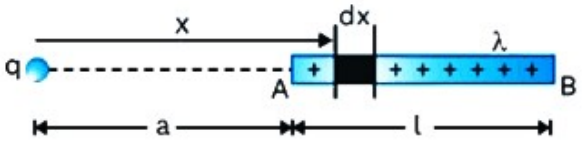
$$F_{net} = 2F\cos\beta$$

$$= \frac{2KQq}{\left(\frac{\sqrt{3}\ell}{2}\right)^2}\left(\frac{\ell/2}{\frac{\sqrt{3}\ell}{2}}\right) \quad F_{net} = \frac{8KQq}{3\sqrt{3}\ell^2}$$

Q. **Find the net force on point charge 'q' due to uniformly charged rod AB with linear charge density λ.**



Sol: The Coulomb's law cannot be directly applied between the total charge on rod and the point charge as Coulomb's law is valid for point charges only.



So, we consider an infinitesimal charge dq of linear width dx on the rod at a distance x from the point charge.

Charge on length 'dx' = dq

$dq = \lambda.dx$

Now, the force on charge element,

$$dF = \frac{KqdQ}{x^2} = \frac{Kq\lambda dx}{x^2}$$

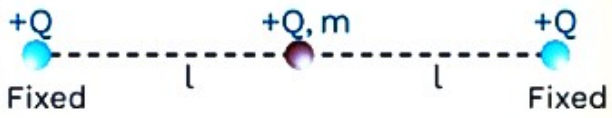
The total force can be calculated by integrating above expression within proper limits.

$$\Rightarrow F = Kq\lambda \int_a^{\ell+a} \frac{dx}{x^2}$$

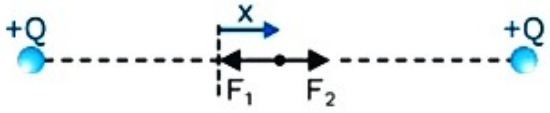
$$\Rightarrow F = Kq\lambda \left[\frac{-1}{x}\right]_a^{\ell+a} \Rightarrow F = Kq\lambda \left[\frac{-1}{\ell+a} + \frac{1}{a}\right]$$

$$\Rightarrow F = \frac{Kq\lambda\ell}{a(a+\ell)}$$

Q. **Find the time period of small oscillations of the central charge along the line joining two charges.**



Sol: On giving a small displacement x to the central charge along the line joining the two charges, the repulsive force on the central charge increases slightly due to one charge while it decreases slightly due to the other charge. Let the forces be F_1 and F_2 as shown in the figure below:



At this displacement, the net force, $F_{net} = F_1 - F_2$

$$F_{net} = \frac{KQ^2}{(\ell - x)^2} - \frac{KQ^2}{(\ell + x)^2} = KQ^2\left[\frac{(\ell + x)^2 - (\ell - x)^2}{(\ell^2 - x^2)^2}\right]$$

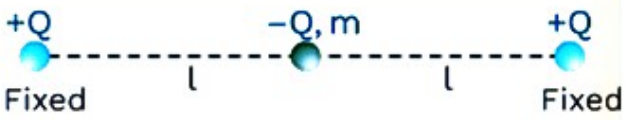
$$F_{net} = KQ^2\left[\frac{4\ell x}{(\ell^2 - x^2)^2}\right] = ma$$

For small oscillation, $\ell >> x$,

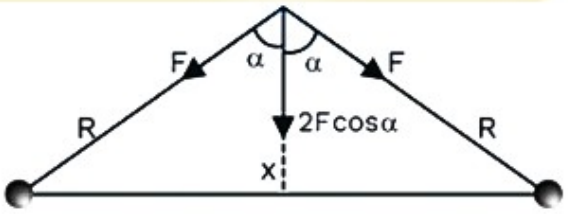
$$\therefore\ ma \approx \frac{4KQ^2\ell}{\ell^4}x \Rightarrow a = \frac{4KQ^2}{\ell^3}x = \omega^2 x$$

$$\Rightarrow \omega = \sqrt{\frac{4KQ^2}{\ell^3}} \Rightarrow T = \frac{2\pi}{\omega} \Rightarrow T = 2\pi\sqrt{\frac{\ell^3}{4\,KQ^2}}$$

Q. **Find the time period of small oscillation of the central charge in direction perpendicular to line joining the two charges.**



Sol: On displacing the central charge slightly by x in the direction perpendicular to the line joining the charges, the forces acting on it are shown in figure below.



Net force on central charge = ma = 2Fcosα

$$a = \frac{2KQ^2}{mR^2}\cdot\frac{x}{R}$$

$$a = \frac{2KQ^2}{mR^3}x = \frac{2KQ^2x}{m(\ell^2 + x^2)^{3/2}}$$

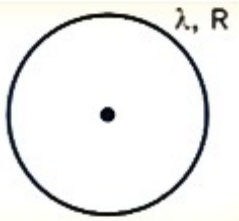
For x<<l,

$$a = \frac{2KQ^2}{m\ell^3}x = \omega^2 x$$

$$\Rightarrow \omega = \sqrt{\frac{2KQ^2}{m\ell^3}}$$

$$T = 2\pi\sqrt{\frac{m\ell^3}{2KQ^2}}$$

Q. **Find the increase in tension in the ring when a charge q is placed at the centre of the uniformly charged ring.**

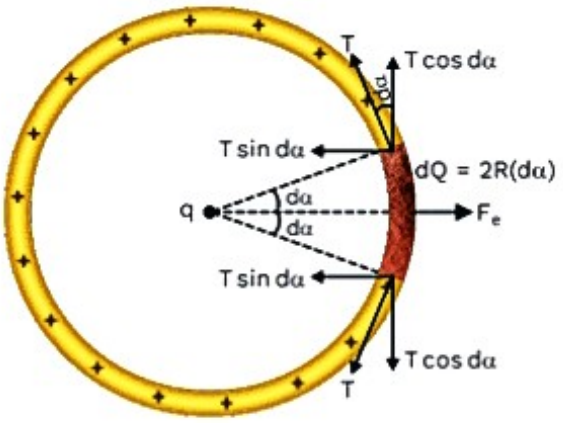


Sol: Consider an element on the periphery of the ring subtending an angle (2. dα) at the centre of the ring as shown in the figure.

The electrostatic force of repulsion between the charge kept at the centre and the charge on the element is shown.

If T is the tension in the ring, then,

In equilibrium,



$$2T\sin(d\alpha) = F_e = \frac{Kq.(dq)}{R^2}$$

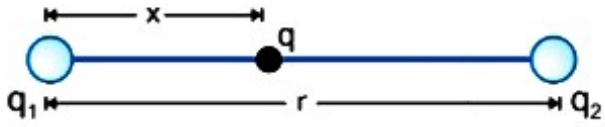
For small angle,

$$2T(d\alpha) = \frac{Kq(2R)(d\alpha)\lambda}{R^2}$$

$$T = \frac{Kq\lambda}{R} = \left(\frac{1}{4\pi\varepsilon_o}\right)\frac{q\lambda}{R}$$

Equilibrium condition for three charged particles

The equilibrium of a charged particle, under the action of single force can never be stable. In equilibrium, net force on each charged particle must be zero.



For stable equilibrium,

If $q_1 > 0$, $q_2 > 0$ then q must be less than 0.

If $q_1 < 0$, $q_2 < 0$ then q must be more than 0.

"There is no other possibility for the system to exist in stable equilibrium."

Net force on charge (q_2) can be written as:

$$\frac{Kq_1q_2}{r^2} + \frac{Kq_1q_2}{(r-x)^2} = 0 \qquad \ldots(1)$$

Net force on charge (q): $\frac{Kq_1q}{x^2} + \frac{Kq_2q}{(r-x)^2} = 0$ ...(2)

After solving these two equations, we will get:

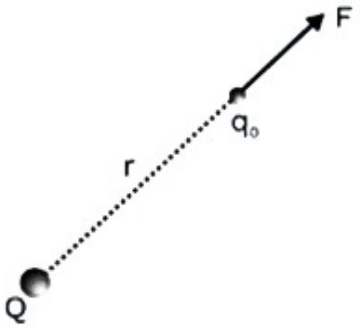
$$\boxed{x = \frac{\sqrt{Q_1}}{\sqrt{Q_1}+\sqrt{Q_2}}(r)} \quad ; \quad \boxed{x = \frac{-Q_1Q_2}{\left(\sqrt{Q_1}+\sqrt{Q_2}\right)^2}}$$

Electric Field

Electric Field Intensity

The space surrounded by a charge in which another charged particle experiences an electrostatic force is known as "electric field" or "electric field intensity".
It is defined as electrostatic force experienced per unit positive charge.

$F = \frac{KQq_0}{r^2}$ = Force experienced by positive charge 'q' due to presence of charge 'Q'.



$$E = \lim_{q_0 \to 0} \frac{F}{q_0} = \frac{KQ\frac{q_0}{r^2}}{q_0} = \frac{KQ}{r^2}$$

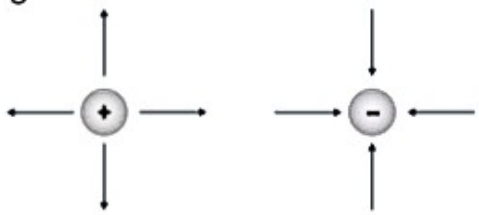
So, Electric field intensity due to charge 'Q' on point where charge 'q_0' is placed is given by,

$$E = \frac{KQ}{r^2}$$

S.I. unit of electric field intensity is $\frac{N}{C}$.

Properties of electric field

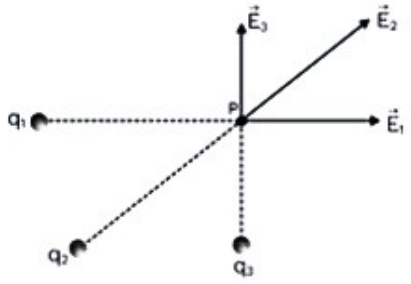
(1) Electric field is a vector quantity. Its direction is same as the direction of force on a positive charge.
(2) Its SI unit is Newton/Coulomb (N/C) or volts/metre (V/m).
(3) Direction of electric field due to positive charge is away from the charge and due to negative charge is towards the charge.



(4) Positive charge experiences electrostatic force $\vec{F}$ in the direction of electric field $\vec{E}$, but negative charge experiences electrostatic force $\vec{F}$ in direction opposite to $\vec{E}$.

$$\vec{E} = \frac{\vec{F}}{q} \Rightarrow \vec{F} = q\vec{E}$$ (Put 'q' with sign)

(5) Electric field follows the principle of superposition.



$\vec{E}_1$ = Electric field at point 'P' due of 'q_1'

$\vec{E}_2$ = Electric field at point 'P' due to 'q_2'

$\vec{E}_3$ = Electric field at point P due to 'q_3'

Then, net electric field at point P will be the vector sum of the electric fields due to q_1, q_2 and q_3.

$$\vec{E}_{net} = \left(\vec{E}_1 + \vec{E}_2 + \vec{E}_3\right)$$

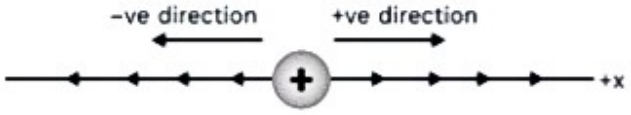
Electric field due to a point charge

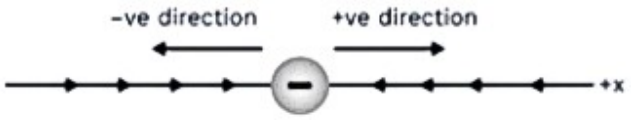
We have already seen that electric field due to a point charge Q at a distance r is given by,

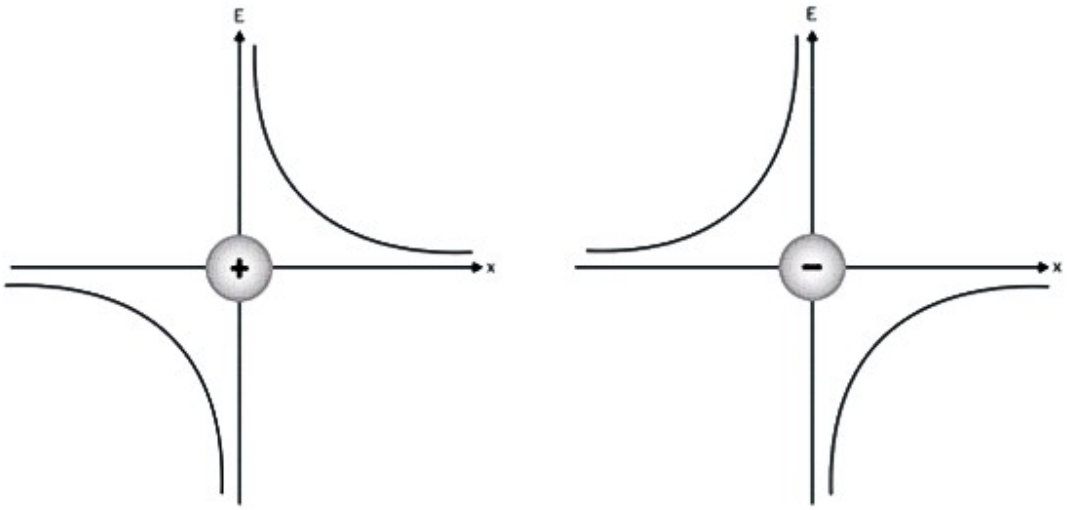
$$E = \frac{KQ}{r^2}$$

In vector form, $\vec{E} = \frac{KQ}{r^2}\hat{r} = \left(\frac{KQ}{r^3}\right)\vec{r}$

As distance of the point (where we want to write the electric field) from the charge (due to which we are writing electric field) increases, i.e., r increases, the value (strength) of electric field decreases.







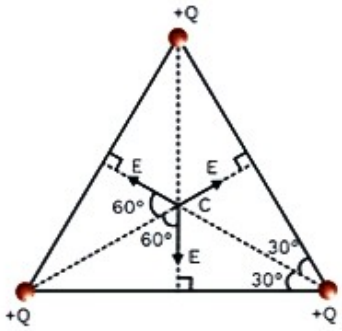
Application of Electric field due to point charge: Regular Polygon

If identical charges are kept at all the vertices of a regular polygon, then net electric field at the centre of polygon is zero.

(a) Equilateral triangle

Electric fields due to the charges kept at the corners are at 120° with each other. Due to symmetry, their vector sum is zero.

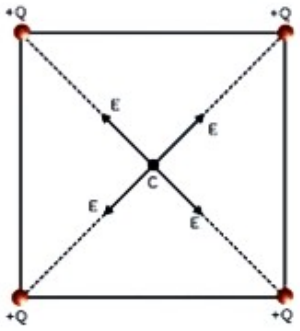
$$(E_c)_{net} = 0$$



(b) Square

Electric fields at the centre due to oppositely placed charges will cancel out and net field at the centre will be zero.

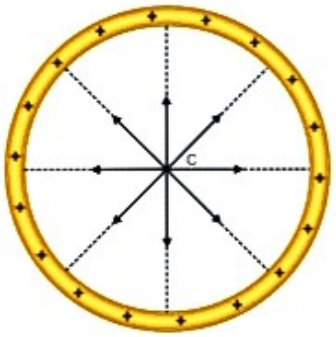
$$(E_c)_{net} = 0$$



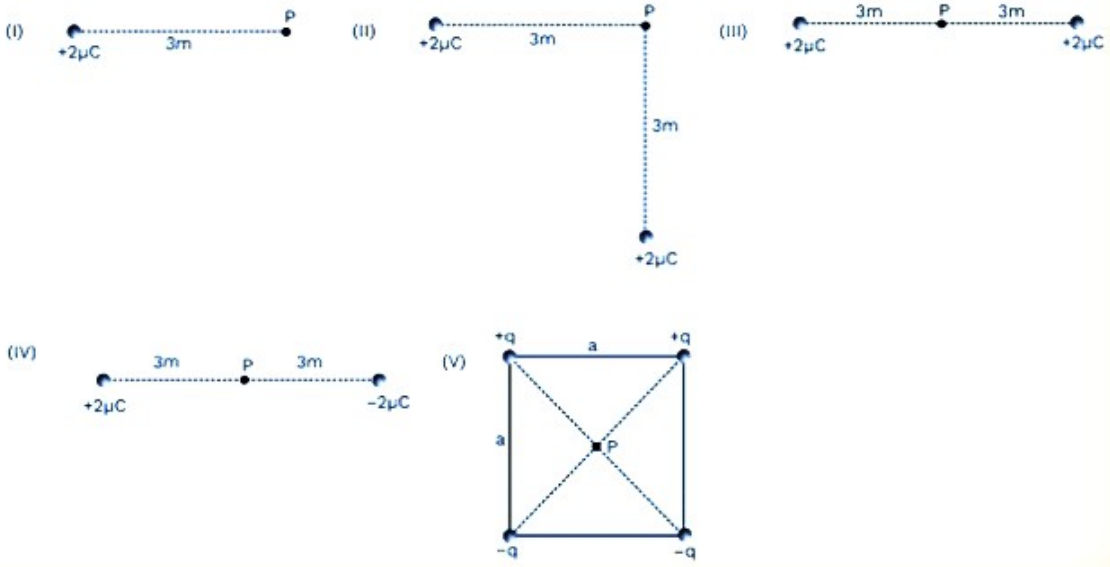
(c) Uniformly charged ring (circle)

Every elemental charge on the periphery of the ring has a counterpart, the fields of which cancel each other, thereby making the net field at the centre of ring to be zero.

$$(E_c)_{net} = 0$$



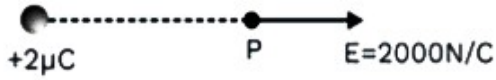
Q. **Find the electric field at point P in the following cases:**



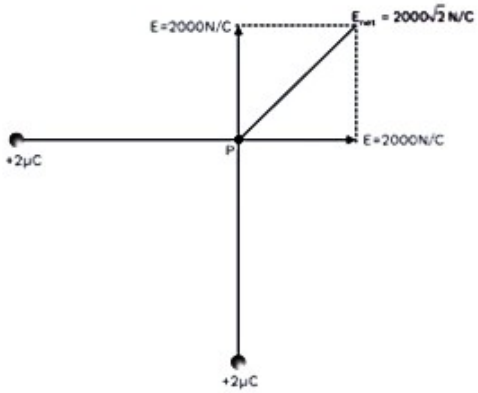
Sol: **(i)** Electric field due to the point charge,

$$E = \frac{KQ}{r^2}$$

$$= \frac{9 \times 10^9 \times 2 \times 10^{-6}}{3^2} = 2000 NC^{-1}$$



(ii) Value of E at P due to individual charges at P is 2000 N/C. But as there are two fields separately produced at point 'P', so to find the resultant, superposition principle is used.

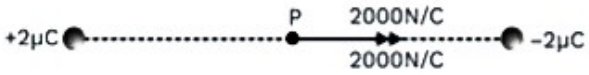


(iii) Electric field due to each positive charge is away from it having same magnitude. As the fields are equal and opposite to each other, the net field at point P is zero.

$E_p = 0$

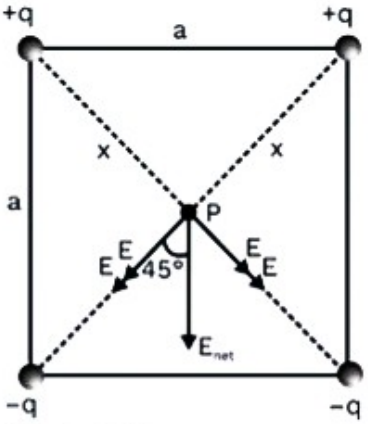


(iv) Electric field due to positive charge is away from it while the electric field due to negative charge is directed towards it. The magnitude and the direction of electric fields due to both the charges are same as shown in the figure.



Net electric field at point P, $E_p = 2000 + 2000 = 4000$ N/C (towards the -2 μC charge).

(v) The electric fields at centre of square due to the positive charges kept on the corners are away from them while the fields due to negative charges are towards them.



From the figure, $x^2 + x^2 = a^2 \Rightarrow x = \frac{a}{\sqrt{2}}$

Electric field magnitude at P due to a single charge,

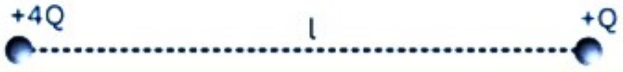
$$E = \frac{Kq}{(x)^2} = \frac{Kq}{\left(\frac{a}{\sqrt{2}}\right)^2} = \frac{2Kq}{a^2}$$

Now, net electric field at P will be the resultant of 2E and 2E at 90°.

$$E_{net} = \sqrt{(2E)^2 + (2E)^2} = 2\sqrt{2}E = 2\sqrt{2} \times \frac{2Kq}{a^2} = \frac{4\sqrt{2}Kq}{a^2}$$

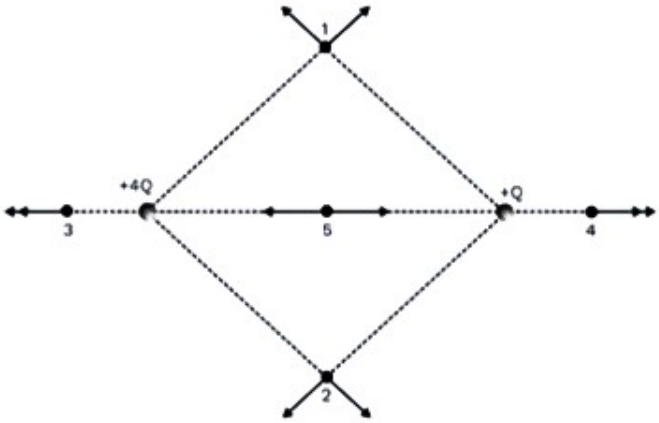
(the direction is downwards).

Q. **Find the point(s) where electric field is zero due to the charges shown.**



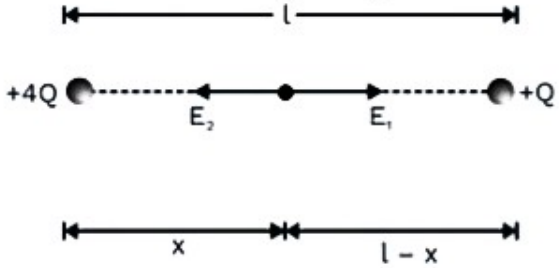
Sol: Lets locate some points 1, 2, 3, 4 and 5 around the charges as shown in the figure. We can do a qualitative analysis to find the points where electric field can be possibly zero.

Electric field due to positive charges are away from them. So, mark electric field vectors due to both the charges at these points.



Electric field cannot be zero at points 1, 2, 3 and 4 as these vector resultants will never be zero. It can be zero at point '5' only if magnitude of fields produced by individual charges are equal in magnitude.

Now, we can find the exact location of point 5 as described below. Let the distance of this point be x from the +4Q charge.



For net electric field at this point to be zero, $E_1 = E_2$

$$\frac{K(4Q)}{x^2} = \frac{KQ}{(\ell - x)^2}$$

$$4(\ell - x)^2 = x^2$$

$$2(\ell - x) = \pm x$$

Considering the positive result, $2(\ell - x) = x$

$$x = \frac{2\ell}{3}$$

Considering the negative result, $2(\ell - x) = -x$

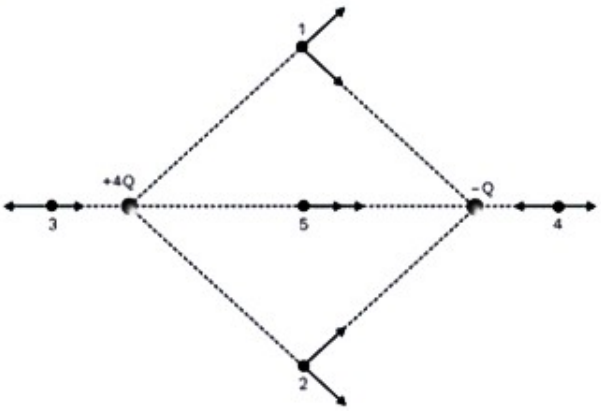
$x = 2\ell$ .... (not possible as the separation is ℓ only)

So, the point where electric field is zero lies at a distance of $\frac{2\ell}{3}$ from 4Q or at distance $\frac{\ell}{3}$ from Q.

Q. **Find the points where electric field is zero due to the charges shown.**



Sol: Repeating the method from previous question by marking points 1, 2, 3, 4 and 5 as shown and marking the electric fields at these points by both the charges.



We can say directly that electric field cannot be zero at points 1, 2 and 5.

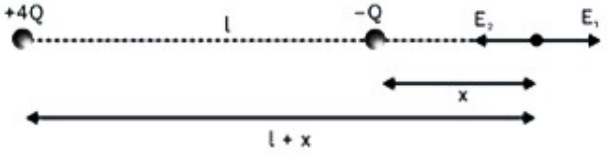
At point 3, the magnitude of field due to '4Q' will be more as compared to the field due to '–Q' because its charge is more and distance of point 3 from charge '4Q' is less.

So, from qualitative analysis, we can say that electric field can be zero at point '4'. Let's consider this point lies at a distance x from -Q as shown.

Now, for electric field to be zero at point 4,

$$E_1 = E_2$$

$$\frac{K(4Q)}{(\ell + x)^2} = \frac{Kq}{x^2}$$



$$\Rightarrow 4x^2 = (\ell + x)^2 \Rightarrow \pm 2x = \ell + x$$

Considering the positive result,

$$+2x = \ell + x \Rightarrow x = \ell \Rightarrow x = \ell$$

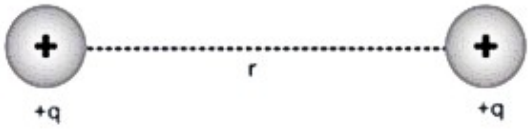
Considering the negative result,

$$-2x = \ell + x \Rightarrow x = -\frac{\ell}{3}$$ .... (Negative value for x is not possible).

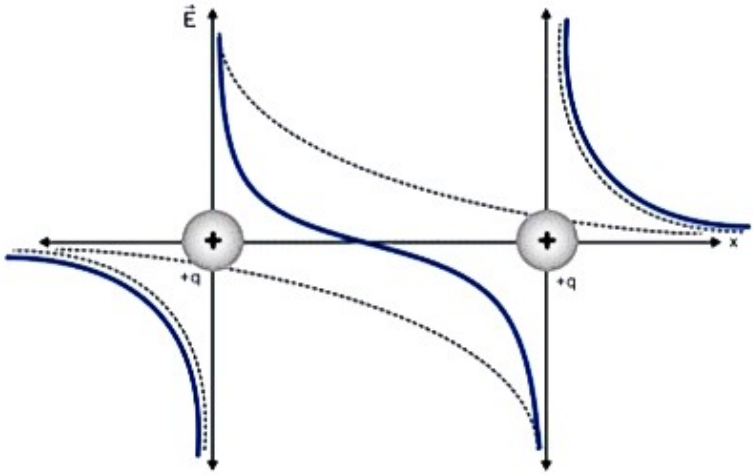
So, electric field is zero at a distance ℓ from '–Q' and 2ℓ from '+4Q'.

Drawing the graph for electric field variation on a line joining two point charges

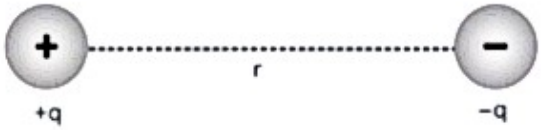
Consider two point charges of same magnitude and nature separated by a distance r. We have to draw a graph for the variation of the electric field along the line joining the charges.



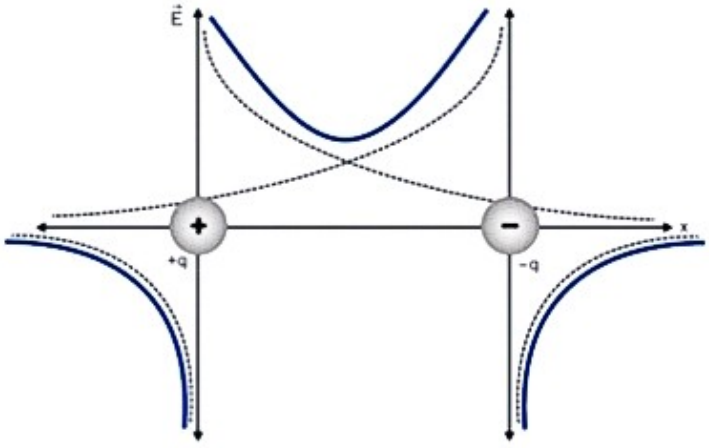
Step-1 : Draw the E-x graph for individual charges.
Step-2 : Superimpose both the fields to find the resultant field.



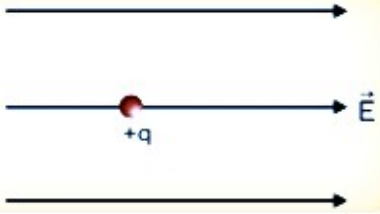
Now consider two point charges of same magnitude but opposite nature separated by a distance r.



Step-1 : Draw the E-x graph for individual charges.
Step-2 : Superimpose both the fields to find the resultant field.

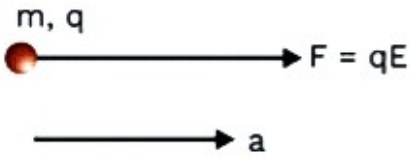


Q. **A particle of mass 'm' and charge '+q' is placed in a uniform electric field. Find acceleration, velocity, and displacement of the particle at time 't'. Initial velocity of the particle is zero.**



Sol: Force on the charge, $F = qE = ma$

$\Rightarrow qE = ma \Rightarrow a = \frac{qE}{m}$



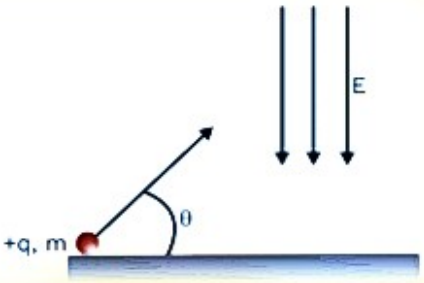
As the acceleration is constant, we can use the equations of motion for the charged particle.

Velocity, $v = 0 + at \Rightarrow v = \frac{qE}{m}t$

Displacement, $S = \frac{1}{2}at^2$

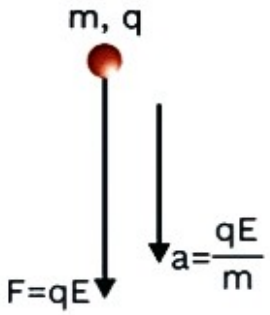
$$S = \frac{1}{2}\frac{qE}{m}t^2$$

Q. **A particle of mass 'm' and charge '+q' is projected at an angle 'θ' with speed 'u' as shown in the figure in a uniform downward electric field. Neglecting gravity, find the time of flight, maximum height and range of the particle.**



Sol: This problem is similar to the projectile motion under gravity, just replace 'g' by $\frac{qE}{m}$.

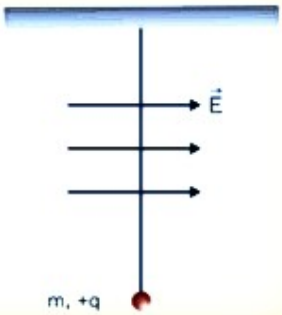
Time of flight, $T = \frac{2u\sin\theta}{g} = \frac{2u\sin\theta}{\left(\frac{qE}{m}\right)} = \frac{2mu\sin\theta}{qE}$



Maximum height, $H = \frac{u^2\sin^2\theta}{2g} = \frac{mu^2\sin^2\theta}{2qE}$

Range, $R = \frac{u^2\sin 2\theta}{g} = \frac{mu^2\sin 2\theta}{qE}$

Q. **A particle of mass 'm' and charge '+q' is suspended by a thread as shown in the figure. A uniform electric field is switched on at t = 0. Find the angle between thread and vertical in equilibrium condition. Find the time period for small oscillation about the equilibrium position. Also, find maximum angle between the thread and vertical after electric field is switched on.**



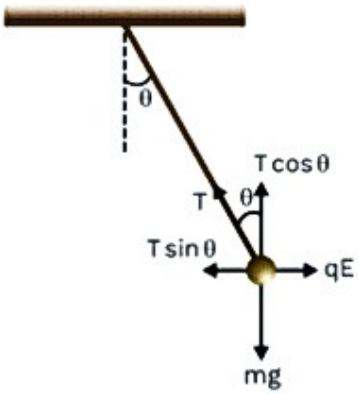
Sol: In equilibrium position of the charge,

$T\sin\theta = qE$

$T\cos\theta = mg$

$\tan\theta = \frac{qE}{mg}$

$\Rightarrow$ Angle at equilibrium position, $\theta_{eq} = \tan^{-1}\left(\frac{qE}{mg}\right)$

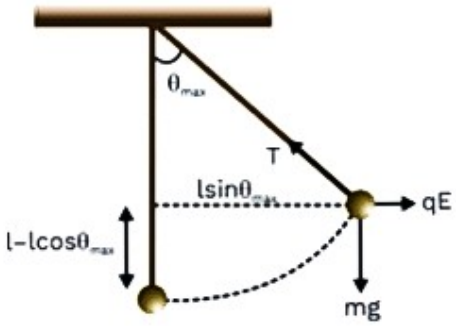


Now, at this equilibrium position, $g_{eff} = \frac{\sqrt{(mg)^2 + (qE)^2}}{m}$

Time period, $T = 2\pi\sqrt{\frac{\ell}{g_{eff}}}$

Let maximum angle between the thread and vertical is 'θ_{max}'.

Applying work energy theorem,



$W_{gravity} + W_{Tension} + W_{Electrostatic} = K_f - K_i$

$-mg\ell(1-\cos\theta) + 0 + qE\ell\sin\theta = 0 - 0$

$qE\sin\theta = mg(1-\cos\theta)$

$qE\left(2\sin\frac{\theta}{2}\cos\frac{\theta}{2}\right) = mg\left[2\sin^2\frac{\theta}{2}\right]$

$\tan\frac{\theta}{2} = \frac{qE}{mg}$

$\theta_{max} = 2\tan^{-1}\left(\frac{qE}{mg}\right)$

Q. **Find electric field at the point $(3\hat{i}+4\hat{j})$ m due to a point charge 5C placed at origin.**

Sol: We know that electric field due to a point charge in vector form is given by,

$$\vec{E}=\frac{Kq\vec{r}}{r^3}$$

$$=\frac{9\times10^9\times5}{\left[\left(3^2+4^2\right)^{1/2}\right]^3}\left(3\hat{i}+4\hat{j}\right)$$

$$=\frac{9\times10^9\times5}{125}\left(3\hat{i}+4\hat{j}\right)$$

$$\vec{E}=36\times10^7\left(3\hat{i}+4\hat{j}\right)N/C$$

Q. **If the force experienced by a +2C charge due to an electric field is $(6\hat{i}+10\hat{j})$ N, then find the force experienced by –3C charge due to the same electric field.**

Sol: We know that force acting on a charge q due to an electric field is given by,

$$\vec{F}=q\vec{E}$$

$$\Rightarrow 6\hat{i}+10\hat{j}=2\vec{E}$$

$$\Rightarrow \vec{E}=\left(3\hat{i}+5\hat{j}\right)NC^{-1}$$

$$\vec{F}_2=q_2\vec{E}=(-3)\left(3\hat{i}+5\hat{j}\right)$$

$$\vec{F}_2=\left(-9\hat{i}-15\hat{j}\right)N.$$

Electric field due to uniform line charge at a point on its axis

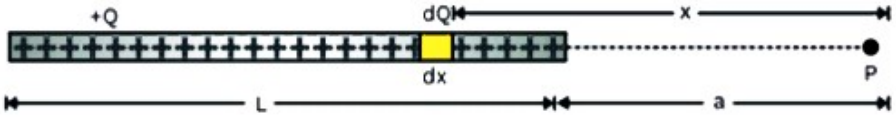
Consider a rod of length L having a charge +Q uniformly distributed over its length. A point P lies on the axis of the rod at a distance "a" from its one end as shown in the figure.



As charge is uniformly distributed, linear charge density can be calculated by unitary method.

Linear charge density, $\lambda = \frac{Q}{L}$

Charge present on very small length dx, $dQ = \lambda dx = \frac{Q}{L}dx$



Electric field at point 'P' due to dQ charge, $dE_p = \frac{K\,dQ}{x^2}$

$$\Rightarrow dE_p = \frac{K}{x^2}\left(\frac{Q}{L}dx\right) = \frac{KQ}{L}\frac{dx}{x^2}$$

To find resultant electric field at P, we have to add electric fields due to all 'dQ' charges.

$$\therefore \text{ Net electric field at P, } E_p = \int dE_p = \int \frac{KQ}{L}\,\frac{dx}{x^2}$$

$$\Rightarrow \quad E_p = \frac{KQ}{L}\int_a^{a+L}\frac{dx}{x^2}$$

$$E_p = \frac{KQ}{L}\left[-\frac{1}{x}\right]_a^{a+L}$$

$$E_p = \frac{KQ}{L}\left[\frac{1}{a} - \frac{1}{a+L}\right] = \frac{KQ}{L}\left[\frac{a+L-a}{a(a+L)}\right] = \frac{KQ}{a(a+L)}$$

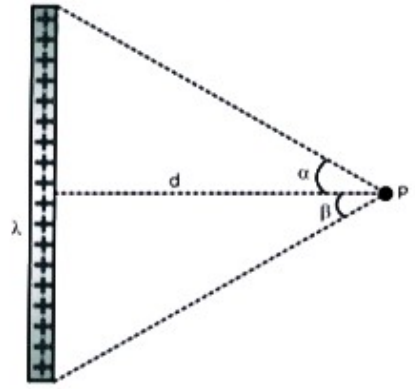
If $a >> L \Rightarrow a + L \approx a$

$$\therefore E_p \approx \frac{KQ}{a^2}$$

Rod will behave like a point charge for a very distant point.

Electric field due to a line charge at a general point

Consider a point P at a perpendicular distance "d" from a uniformly charged rod making angles α and β as shown.



Linear charge density, $\lambda = \frac{dQ}{dy}$

Consider an elemental charge on rod at a distance y as shown.

$dQ = \lambda.dy$

Electric field at point P due to 'dQ' charge,

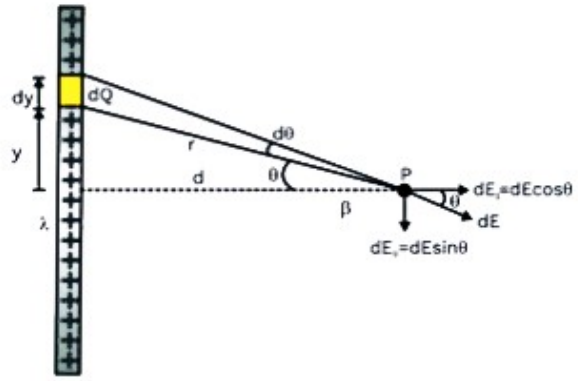
$$dE = \frac{K(\lambda dy)}{r^2}$$

Now, $\tan\theta = y/d \Rightarrow y = d\tan\theta$

$$\therefore \frac{dy}{d\theta} = d \times \sec^2\theta \quad \Rightarrow dy = d \times \sec^2\theta\, d\theta$$

$$\cos\theta = \frac{d}{r} \quad \Rightarrow r = \frac{d}{\cos\theta} = d \times \sec\theta$$

$$\therefore dE = \frac{K\lambda\, dy}{r^2} = \frac{K\lambda\, d \times \sec^2\theta\, d\theta}{d^2 \sec^2\theta} = \left(\frac{K\lambda}{d}\right) d\theta$$



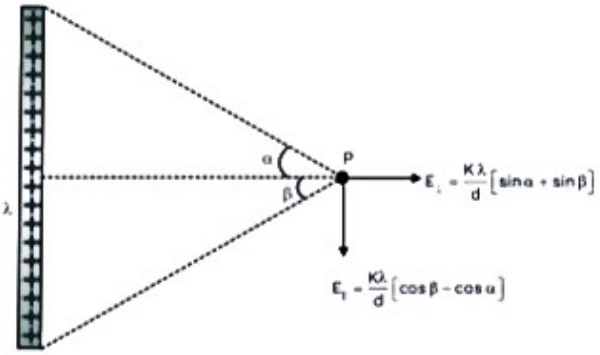
Resultant electric field at point 'P' cannot be calculated by integrating dE, because direction of electric field 'dE' due to each 'dQ' charge will be different. Hence, we will calculate resultant field by component method.

$$dE_\perp = dE\cos\theta = \frac{K\lambda}{d}\cos\theta\, d\theta$$

$$\therefore \quad E_\perp = \int dE_\perp = \frac{K\lambda}{d}\int_{-\beta}^{\alpha}\cos\theta\, d\theta$$

$$E_\perp = \frac{K\lambda}{d}\left[\sin\theta\right]_{-\beta}^{\alpha} = \frac{K\lambda}{d}\left[\sin\alpha - \sin(-\beta)\right]$$

$$\Rightarrow \quad \boxed{E_\perp = \frac{K\lambda}{d}\left[\sin\alpha + \sin\beta\right]}$$



Now, $dE_{\parallel} = dE\sin\theta = \frac{K\lambda}{d}\sin\theta\, d\theta$

$$\therefore E_{\parallel} = \int dE_{\parallel} = \frac{K\lambda}{d}\int_{-\beta}^{\alpha}\sin\theta d\theta$$

$$E_{\parallel} = \frac{K\lambda}{d}\left[-\cos\theta\right]_{-\beta}^{\alpha} = \frac{K\lambda}{d}\left[\cos(-\beta) - \cos\alpha\right]$$

$$\Rightarrow \boxed{E_{\parallel} = \frac{K\lambda}{d}\left[\cos\beta - \cos\alpha\right]}$$

Net electric field, $E_{net} = \sqrt{E_{\perp}^2 + E_{\parallel}^2}$

Now, let's find the direction of net electric field at 'P'. Let net electric field makes an angle ϕ with $E_{\perp}$.

$$\tan\phi = \frac{E_{\parallel}}{E_{\perp}}$$

$$= \frac{(K\lambda/d)[\cos\beta - \cos\alpha]}{(K\lambda/d)[\sin\alpha + \sin\beta]} = \frac{\cos\beta - \cos\alpha}{\sin\alpha + \sin\beta}$$

$$\Rightarrow \tan\phi = \frac{2\sin\left(\frac{\alpha+\beta}{2}\right)\sin\left(\frac{\alpha-\beta}{2}\right)}{2\sin\left(\frac{\alpha+\beta}{2}\right)\cos\left(\frac{\alpha-\beta}{2}\right)} = \frac{\sin\left(\frac{\alpha-B}{2}\right)}{\cos\left(\frac{\alpha-B}{2}\right)}$$

$$\Rightarrow \quad \tan\phi = \tan\left(\frac{\alpha-\beta}{2}\right)$$

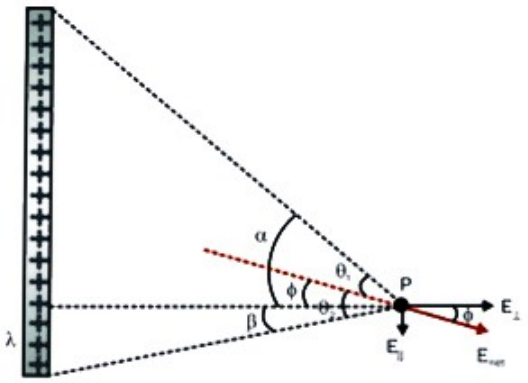
$$\Rightarrow \quad \phi = \left(\frac{\alpha-\beta}{2}\right)$$

Now, $\theta_1 = (\alpha - \phi) = \alpha - \left(\frac{\alpha-\beta}{2}\right) = \alpha - \frac{\alpha}{2} + \frac{\beta}{2}$

$$\Rightarrow \quad \theta_1 = \left(\frac{\alpha+\beta}{2}\right)$$

and, $\theta_2 = \beta + \phi = \beta + \left(\frac{\alpha-\beta}{2}\right) = \left(\frac{\alpha+\beta}{2}\right)$

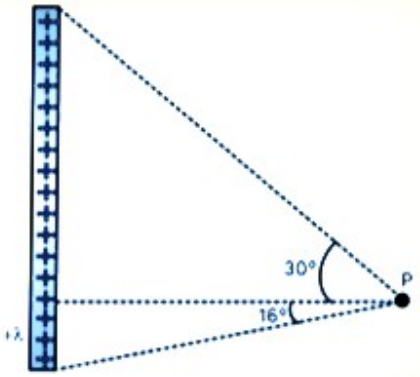
$$\Rightarrow \quad \theta_1 = \theta_2 = \left(\frac{\alpha+\beta}{2}\right)$$



Tip angle = Angle made by rod at point 'P' = $\alpha + \beta$.

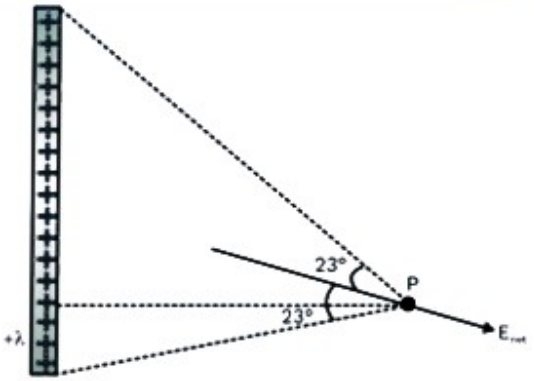
$\therefore$ It can be noted that, net electric field will act on angle bisector line of tip angle at point 'P'.

Q. **Find direction of electric field at point 'P' in the figure where rod is uniformly charged.**



Sol: Direction of the electric field will be along the angle bisector of the tip angle.

$$\frac{\alpha+\beta}{2}=\frac{30^{\circ}+16^{\circ}}{2}=23^{\circ}$$



Special Cases

Case-1: If wire (rod) is infinitely long.

For infinitely long rod, $\alpha = 90°$, $\beta = 90°$

$$E_{\perp}=\frac{K\lambda}{d}[\sin\alpha+\sin\beta]=\frac{K\lambda}{d}\left[\sin 90^{\circ}+\sin 90^{\circ}\right]$$

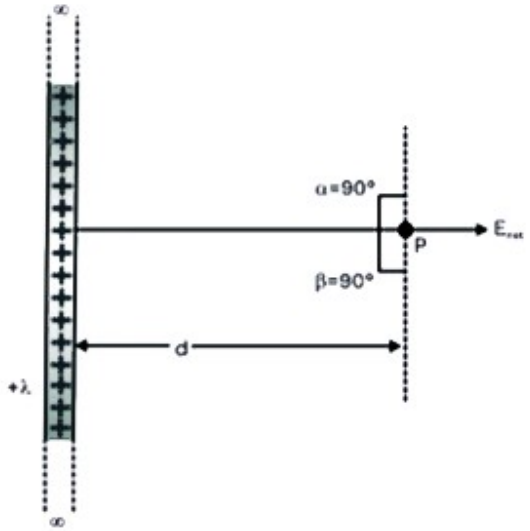
$$E_{\perp}=\frac{2K\lambda}{d}$$

$$E_{\parallel}=\frac{K\lambda}{d}[\cos\beta-\cos\alpha]=\frac{K\lambda}{d}[\cos 90^{\circ}-\cos 90^{\circ}]$$

$$E_{\parallel}=0$$

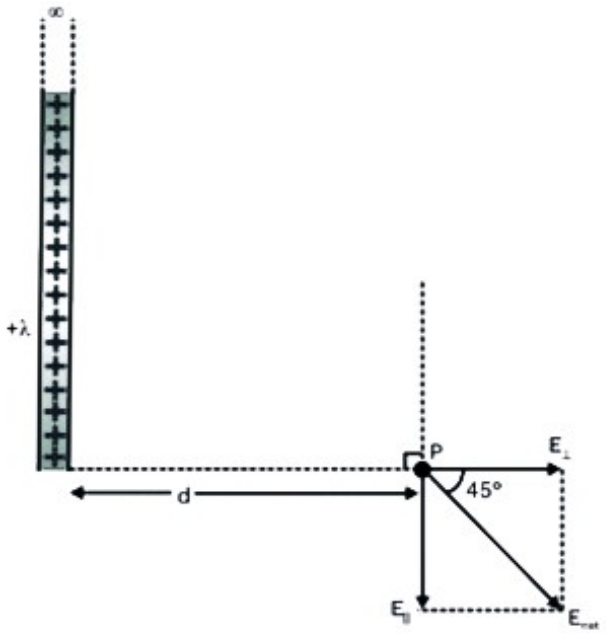
$$\therefore \quad E_{net}=\sqrt{E_{\perp}^{2}+E_{\parallel}^{2}}=E_{\perp}$$

$E_{net}=\frac{2K\lambda}{d}$, direction is along $E_{\perp}$.



Direction of E_{net} can also be calculated by angle bisector concept.

Case-2: If wire (rod) is semi-infinite.



$\alpha = 90°, \beta = 0°$

$$E_{\perp} = \frac{K\lambda}{d}\left[\sin 90° + \sin 0°\right]$$

$$E_{\perp} = \frac{K\lambda}{d}$$

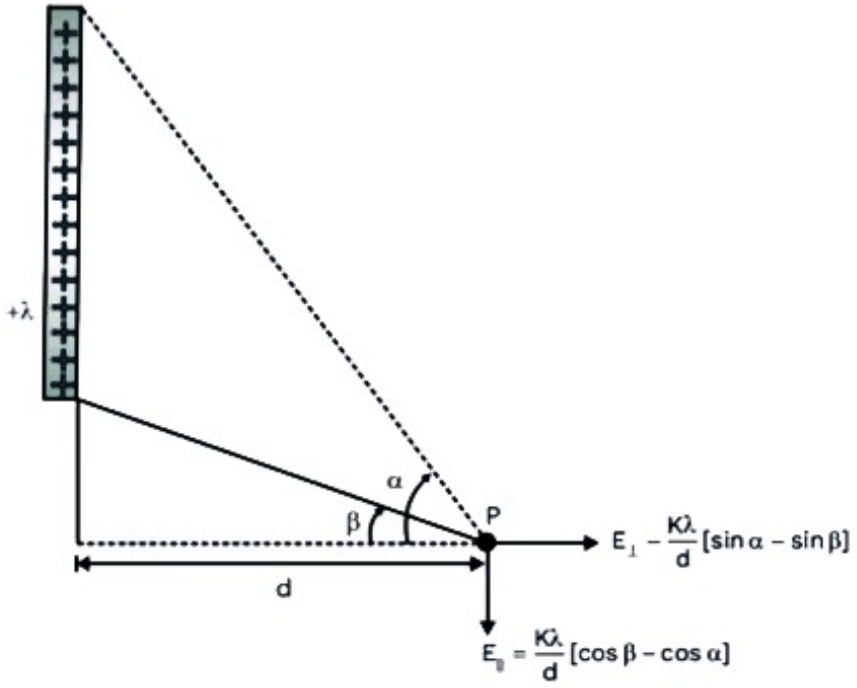
$$E_{\parallel} = \frac{K\lambda}{d}\left[\cos 0° - \cos 90°\right]$$

$$E_{\parallel} = \frac{K\lambda}{d}$$

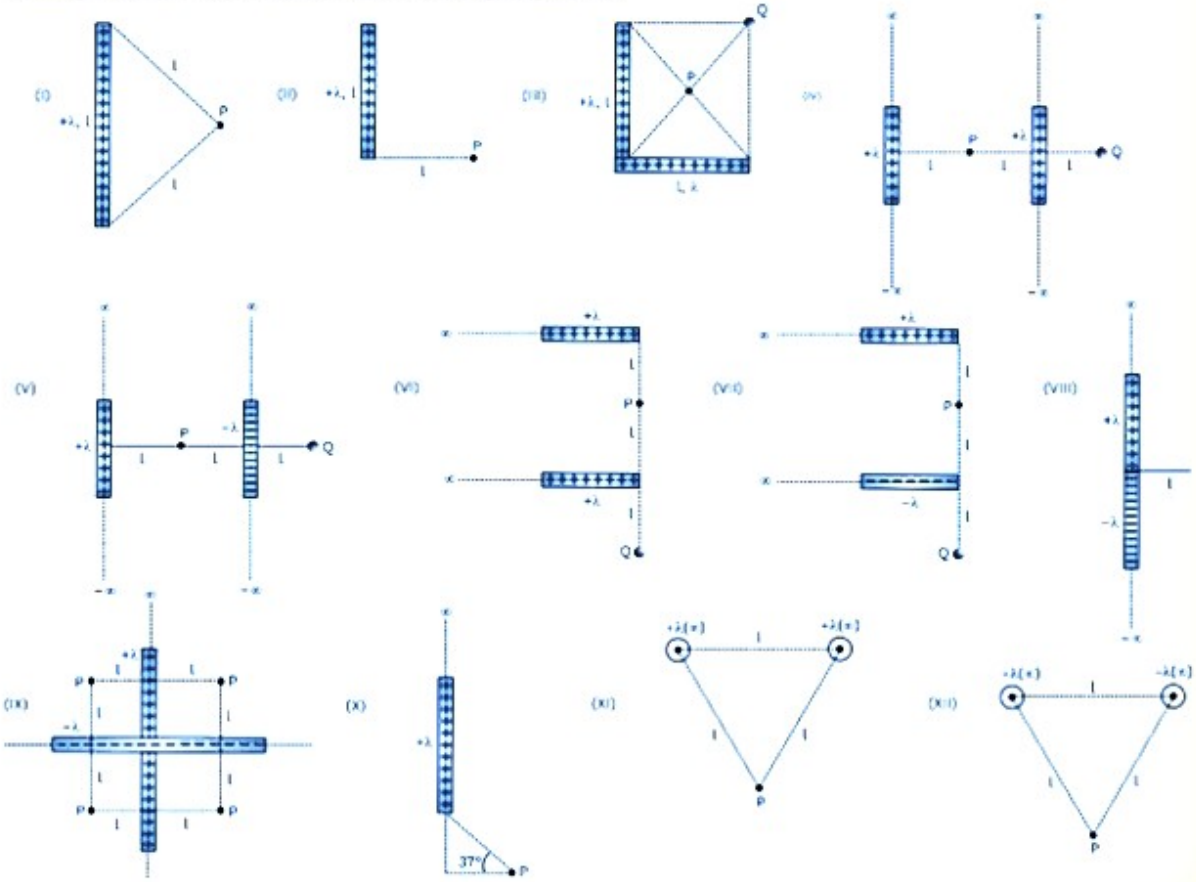
$$E_{net} = \sqrt{\left(\frac{K\lambda}{d}\right)^2 + \left(\frac{K\lambda}{d}\right)^2} = \frac{\sqrt{2}K\lambda}{d}$$

Case 3:

When the point is away from the rod



Q. **Find electric field at point P due to the linear charge distribution and point charge Q (if any) in the following figures:**

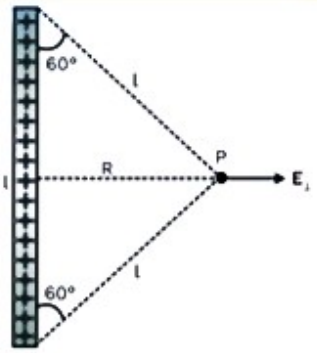


Sol: (I) From figure,

$$R = \ell \sin 60^\circ = \frac{\sqrt{3}\,\ell}{2}$$

$$E_\perp = \frac{K\lambda}{R}\left(\cos 60^\circ + \cos 60^\circ\right) = \frac{2K\lambda}{\sqrt{3}\,\ell}\ , \text{and, } E_\parallel = 0.$$

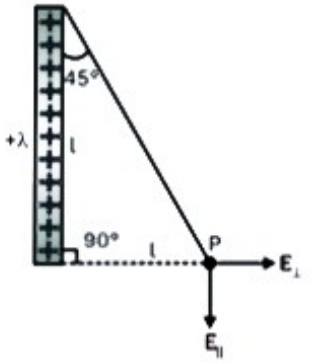
$$E_p = \sqrt{E_\perp^2 + E_\parallel^2} = \frac{2K\lambda}{\sqrt{3}\,\ell}$$



(II) $E_{\perp} = \frac{K\lambda}{\ell}(\cos 90° + \cos 45°) = \frac{K\lambda}{\sqrt{2}\,\ell}$

$E_{\parallel} = \frac{K\lambda}{\ell}(\sin 90° - \sin 45°) = \frac{K\lambda}{\ell}\left(1 - \frac{1}{\sqrt{2}}\right)$

$E_P = \sqrt{E_{\perp}^2 + E_{\parallel}^2} = \frac{K\lambda}{\ell}\sqrt{2 - \sqrt{2}}$

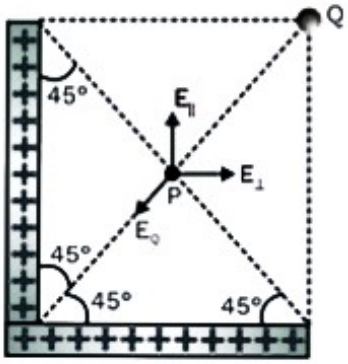


(III) $E_{\perp} = \frac{K\lambda}{(\ell/2)}(\cos 45° + \cos 45°) = \frac{2\sqrt{2}K\lambda}{\ell}$

$E_{\parallel} = \frac{K\lambda}{(\ell/2)}(\cos 45° + \cos 45°) = \frac{2\sqrt{2}K\lambda}{\ell}$

$E_Q = \frac{KQ}{(\ell/\sqrt{2})^2} = \frac{2KQ}{\ell^2}$

$E_P = \sqrt{E_{\perp}^2 + E_{\parallel}^2} - E_Q = \frac{4K\lambda}{\ell} - \frac{2KQ}{\ell^2}$ ... (along PQ)

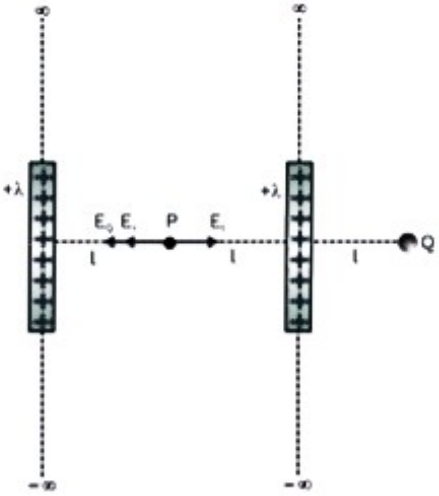


(IV) E_r = Field due to right side line charge = $\frac{2K\lambda}{\ell}$

E_ℓ = Field due to left side line charge = $\frac{2K\lambda}{\ell}$

E_Q = Field due to point charge Q = $\frac{KQ}{(2\ell)^2} = \frac{KQ}{4\ell^2}$

$E_P = E_r + E_Q - E_\ell = E_Q = \frac{KQ}{4\ell^2}$ (Along QP)

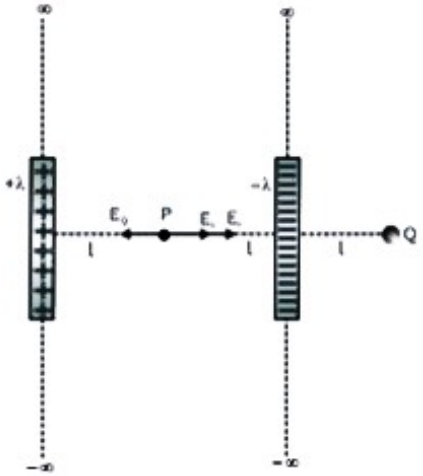


(V) E_r = Field due to right side line charge = $\frac{2K\lambda}{\ell}$

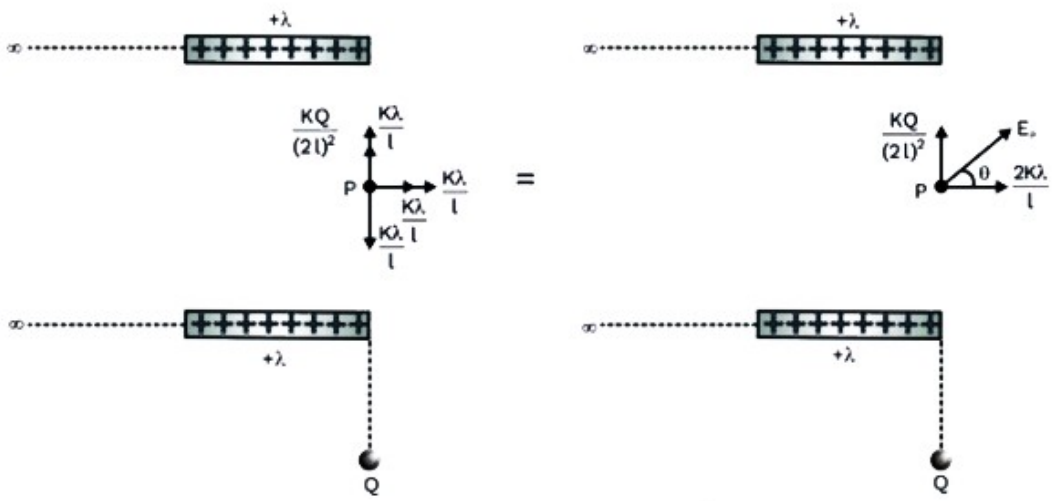
E_ℓ = Field due to left side line charge = $\frac{2K\lambda}{\ell}$

E_Q = Field due to point charge Q = $\frac{KQ}{(2\ell)^2} = \frac{KQ}{4\ell^2}$

$E_P = E_r + E_\ell - E_Q = \frac{4K\lambda}{\ell} - \frac{KQ}{4\ell^2}$ (along PQ)



(VI)



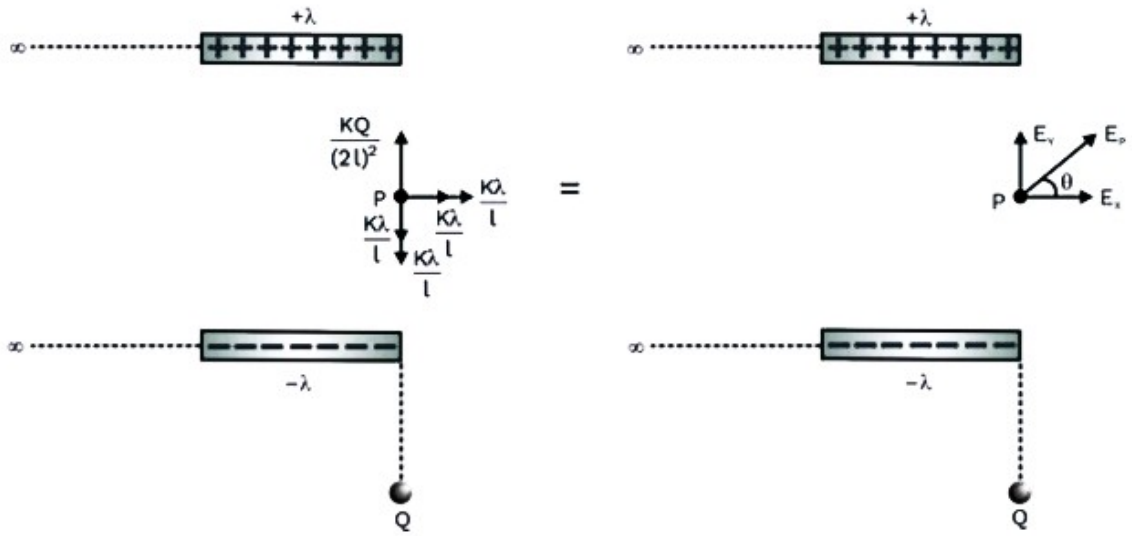
E_X = Field in horizontal direction = $\frac{K\lambda}{\ell} + \frac{K\lambda}{\ell} = \frac{2K\lambda}{\ell}$

E_Y = Field in vertical direction = $\frac{K\lambda}{\ell} + \frac{KQ}{(2\ell)^2} - \frac{K\lambda}{\ell} = \frac{KQ}{4\ell^2}$

$E_P = \sqrt{E_X^2 + E_Y^2} = \sqrt{\left(\frac{2K\lambda}{\ell}\right)^2 + \left(\frac{KQ}{4\ell^2}\right)^2}$ at angle θ from +X-axis as shown in figure.

$$\theta = \tan^{-1}\left(\frac{E_Y}{E_X}\right) = \tan^{-1}\left(\frac{\left(\frac{KQ}{4\ell^2}\right)}{\left(\frac{2K\lambda}{\ell}\right)}\right) = \tan^{-1}\left(\frac{Q}{8\lambda\ell}\right)$$

(VII)



E_x = Field in horizontal direction = $\frac{K\lambda}{\ell} + \frac{K\lambda}{\ell} = \frac{2K\lambda}{\ell}$

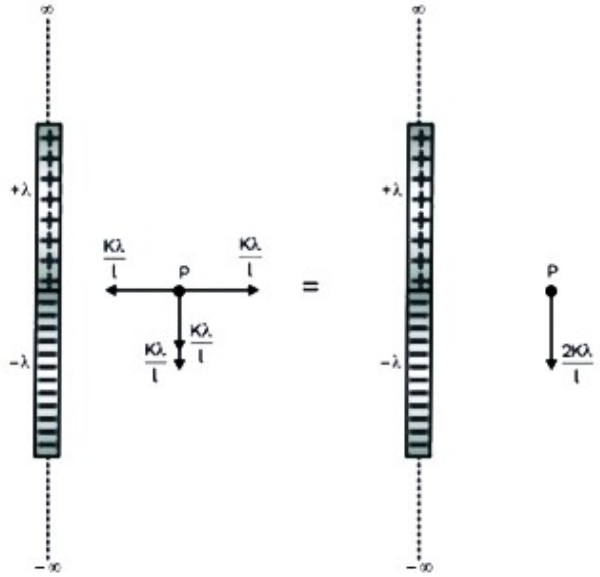
E_y = Field in vertical direction = $\left(\frac{KQ}{(2\ell)^2} - \frac{K\lambda}{\ell} - \frac{K\lambda}{\ell}\right) = \left(\frac{KQ}{4\ell^2} - \frac{2K\lambda}{\ell}\right)$

$E_p = \sqrt{E_x^2 + E_y^2} = \sqrt{\left(\frac{2K\lambda}{\ell}\right)^2 + \left(\frac{KQ}{4\ell^2} - \frac{2K\lambda}{\ell}\right)^2}$ at angle θ from +X-axis as shown in figure.

$\theta = \tan^{-1}\left(\frac{E_y}{E_x}\right) = \tan^{-1}\left(\frac{\frac{KQ}{4\ell^2} - \frac{2K\lambda}{\ell}}{\frac{2K\lambda}{\ell}}\right) = \tan^{-1}\left(\frac{Q}{8\lambda\ell} - 1\right)$

(VIII)

$E_p = \frac{2K\lambda}{\ell}$ (in vertically downward direction)



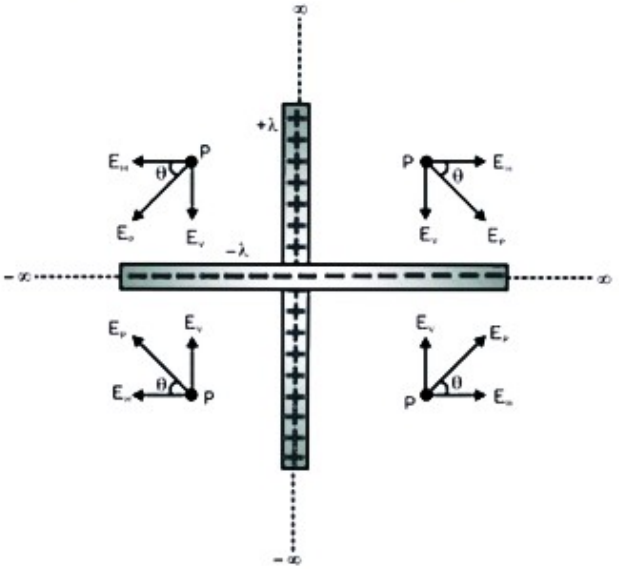
(IX)

E_H = Field in horizontal direction

$= \frac{2K\lambda}{\ell}$

E_V = Field in vertical direction = $\frac{2K\lambda}{\ell}$

$E_p = \sqrt{E_H^2 + E_V^2} = \sqrt{\left(\frac{2K\lambda}{\ell}\right)^2 + \left(\frac{2K\lambda}{\ell}\right)^2} = \frac{2\sqrt{2}K\lambda}{\ell}$



At an angle $\theta = 45°$ as shown in figure.

(X) E_H = Field in horizontal direction

$$= \frac{K\lambda}{\ell}(\cos 0° + \cos 127°) = \frac{K\lambda}{\ell}\left[1 + \left(-\frac{3}{5}\right)\right] = \frac{2K\lambda}{5\ell}$$

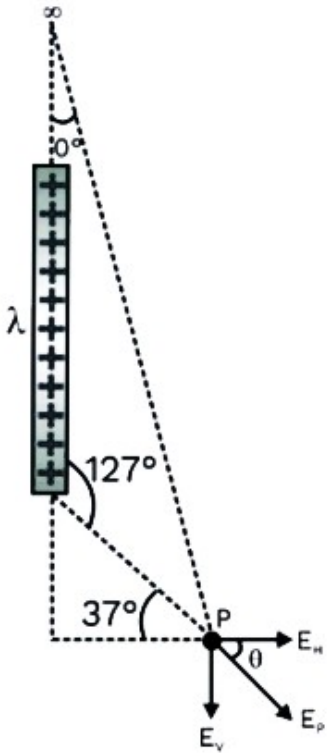
E_V = Field in vertical direction

$$= \frac{K\lambda}{\ell}(\sin 127° - \sin 0°) = \frac{K\lambda}{\ell}\left(\frac{4}{5} - 0\right) = \frac{4K\lambda}{5\ell}$$

$$E_P = \sqrt{E_H^2 + E_V^2} = \sqrt{\left(\frac{2K\lambda}{5\ell}\right)^2 + \left(\frac{4K\lambda}{5\ell}\right)^2} = \frac{2K\lambda}{\sqrt{5}\ell}$$

At angle θ from horizontal direction as shown in figure.

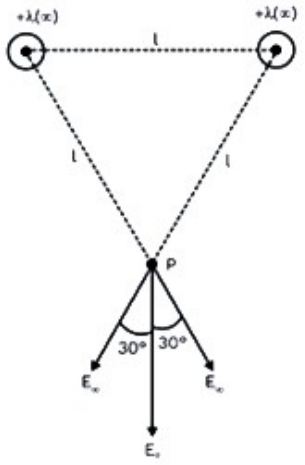
$$\theta = \tan^{-1}\left(\frac{E_V}{E_H}\right) = \tan^{-1}\left(\frac{\frac{4K\lambda}{5\ell}}{\frac{2K\lambda}{5\ell}}\right) = \tan^{-1} 2$$



Wires are perpendicular to the plane of paper in question number '(XI)' & '(XII)'.

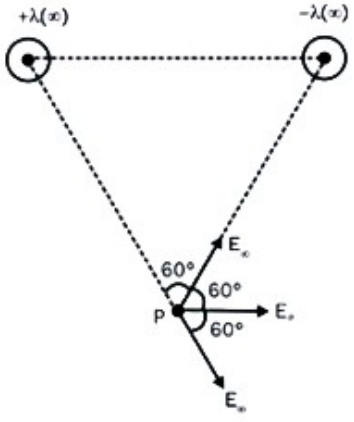
(XI) $E_P = 2E_\infty \cos 30° = 2 \times \frac{2K\lambda}{\ell} \times \frac{\sqrt{3}}{2}$

$$= \frac{2\sqrt{3}K\lambda}{\ell}$$ (in vertically downward direction)

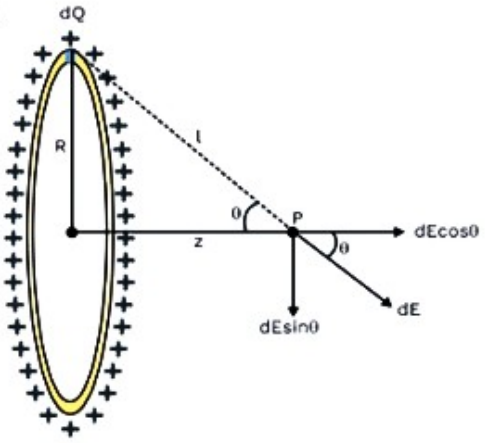


(XII) $E_P = 2E_\infty \cos 60° = 2 \times \frac{2K\lambda}{\ell} \times \frac{1}{2}$

$$= \frac{2K\lambda}{\ell}$$ (in horizontally right direction)



Electric field due to uniformly charged ring on its axis



To find the electric field at a point on the axis of a uniformly charged ring, we consider an infinitesimal charge element on its periphery as shown.

Electric field at point P due to this small charge, $dE = \frac{K\,dQ}{\ell^2}$

where ℓ^2 and $\cos\theta$ are constants.

From the figure,

$\ell = \sqrt{R^2 + z^2}$ and $\cos\theta = \frac{z}{\sqrt{R^2 + z^2}}$

For every small element on the ring, there is another diametrically opposite element whose dE $\sin\theta$ component cancels out the same component of the first element.

For all the elements on the ring, the dE $\cos\theta$ components add up and give the net electric field.

$$E_{axis} = \int dE\cos\theta = \int \frac{KdQ}{\ell^2}\cos\theta$$

$$E_{axis} = \frac{KQ}{\ell^2}\cos\theta = \frac{KQz}{\left(R^2 + z^2\right)^{3/2}}$$

$$E_{\perp \text{ to axis}} = \int dE\sin\theta = 0$$

Net electric field, $E_z = \sqrt{E_{axis}^2 + E_{\perp \text{ to axis}}^2} = E_{axis} = \frac{KQz}{\left(R^2 + z^2\right)^{3/2}}$ (along axis of the ring)

At the location of maximum electric field, $\frac{dE_z}{dz} = 0$.

$$\frac{dE_z}{dz} = KQ\left[\frac{\left(R^2 + z^2\right)^{3/2}\frac{dz}{dz} - z\frac{d}{dz}\left(R^2 + z^2\right)^{3/2}}{\left(R^2 + z^2\right)^3}\right] = 0$$

$$\left(R^2 + z^2\right)\left(R^2 + z^2\right)^{1/2} \times 1 - z \times \frac{3}{2} \times \left(R^2 + z^2\right)^{1/2} \times 2z = 0$$

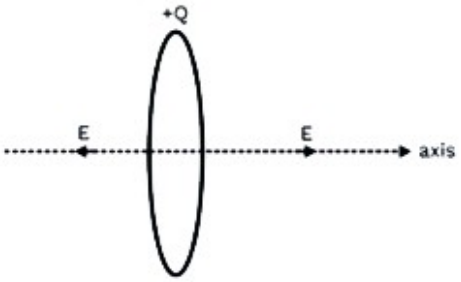
$$R^2 + z^2 - 3z^2 = 0$$

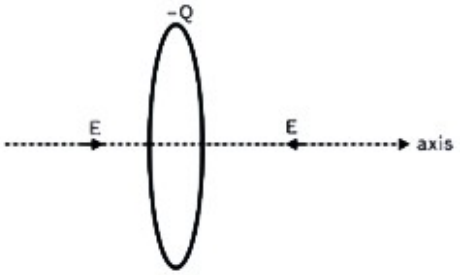
$$2z^2 = R^2$$

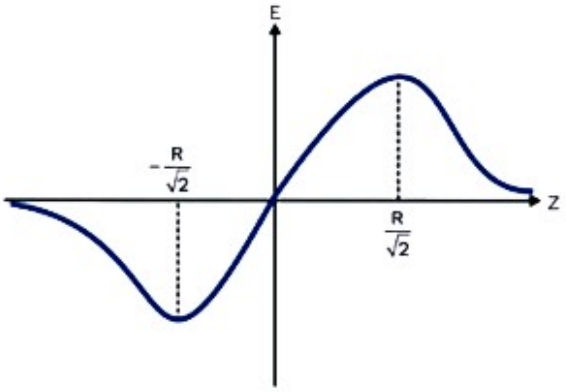
$$\Rightarrow z = \pm\frac{R}{\sqrt{2}}$$

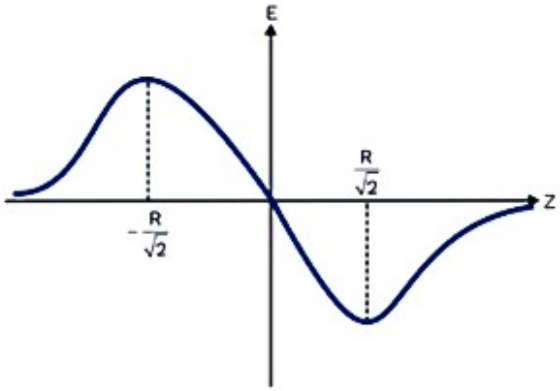
$\therefore$ Electric field on the axis is maximum at $z = \pm\frac{R}{\sqrt{2}}$.

Also, $\left.\frac{dE_z}{dz}\right|_{z=0} \neq 0$, i.e., slope of E-z graph at centre (z = 0) is not equal to zero.



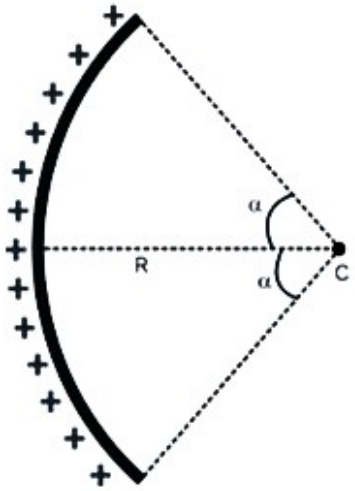






Electric field due to charged circular arc at centre of curvature

Consider a charged circular arc of radius R subtending an angle 2α at the centre of curvature.

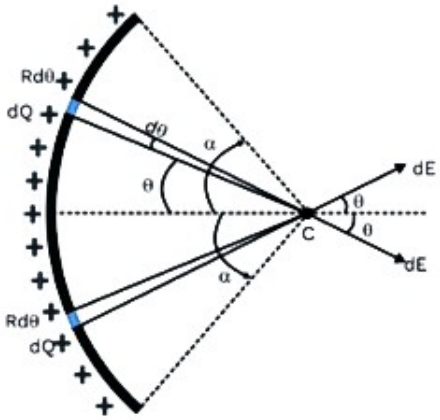


Take an infinitesimal point charge element on arc and write electric field at C due to it. Then add up all the electric fields by taking their components.

$dQ = \lambda R d\theta$

$$dE = \frac{K\,dQ}{R^2} = \frac{K(\lambda R d\theta)}{R^2}$$

We have taken two point-charges '+dQ' as shown. Vertical components of their electric fields (dE sinθ) will cancel out and horizontal components (dE cosθ) will add up.



∴ E_{net} will be along horizontal direction (Angle bisector).

$$E_{net} = \int_{-\alpha}^{\alpha} dE\cos\theta = 2\int_{0}^{\alpha} dE\cos\theta$$

$$= 2\int_{0}^{\alpha} \frac{K\lambda R d\theta}{R^2}\cos\theta = \frac{2K\lambda}{R}\int_{0}^{\alpha}\cos\theta d\theta$$

$$\boxed{E_{net} = \frac{2K\lambda\sin\alpha}{R}}$$

Electric field at the centre of circular arc will act along angle bisector of tip angle.

Special cases

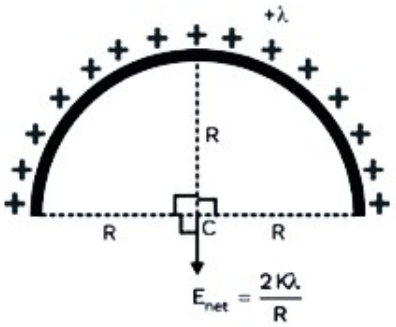
Case-1: Semi-circular arc

For semi-circular arc, $\alpha = 90°$

$$E_{net} = \frac{2K\lambda}{R}\sin 90°$$

$$E_{net} = \frac{2K\lambda}{R}$$

(Along angle bisector of tip angle).



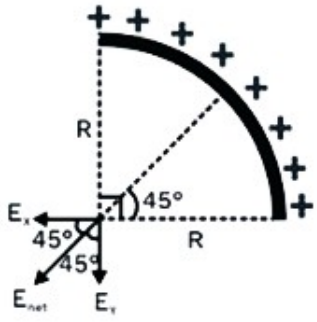
Case-2: Quarter circular arc

For quarter circular arc, $\alpha = 45°$

$$E_{net} = \frac{2K\lambda}{R}\sin 45° = \frac{2K\lambda}{R}\left(\frac{1}{\sqrt{2}}\right)$$

$$E_{net} = \frac{\sqrt{2}K\lambda}{R}$$

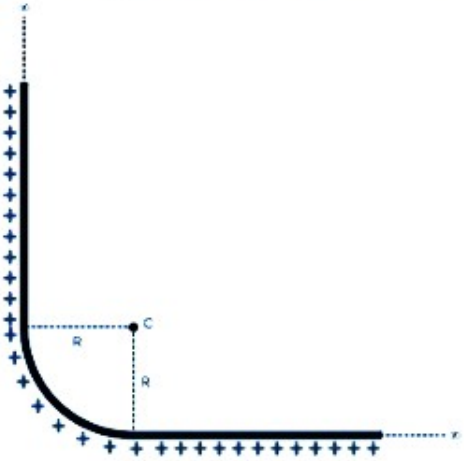
$$E_x = E_{net}\cos 45° = \sqrt{2}\frac{K\lambda}{R}\times\frac{1}{\sqrt{2}} = \frac{K\lambda}{R}$$



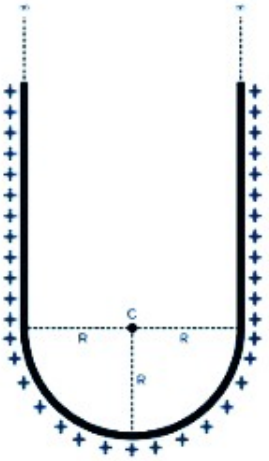
Similarly, $E_y = \frac{K\lambda}{R}$.

Q. **Find the net electric field at point 'C' in the following arrangements. The linear charge density in both the cases is +'λ'.**

(I)

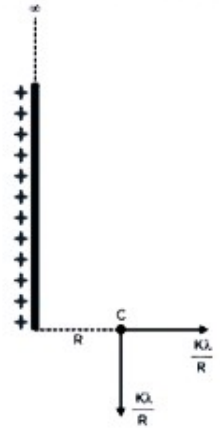


(II)

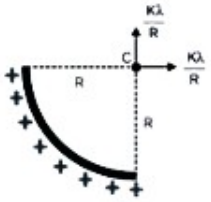


Sol: **(I)** We can break the given arrangement in three portions and write their fields at C. Then calculate resultant field at 'C'.

(a)



(b)

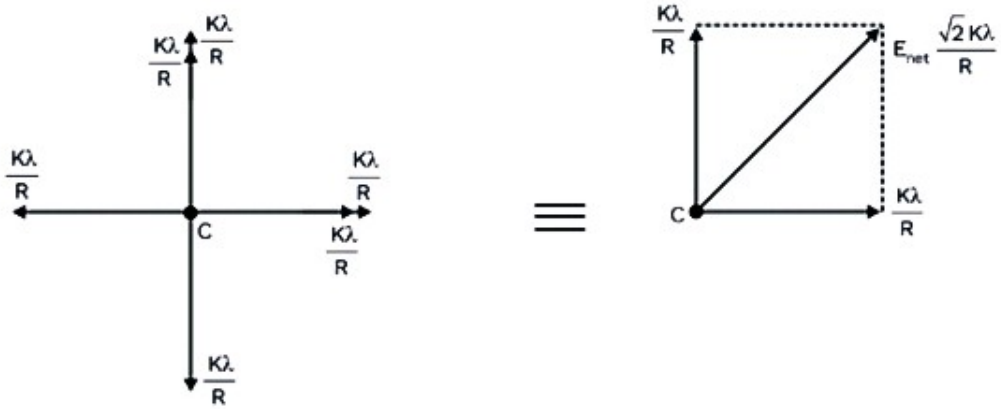


(c)

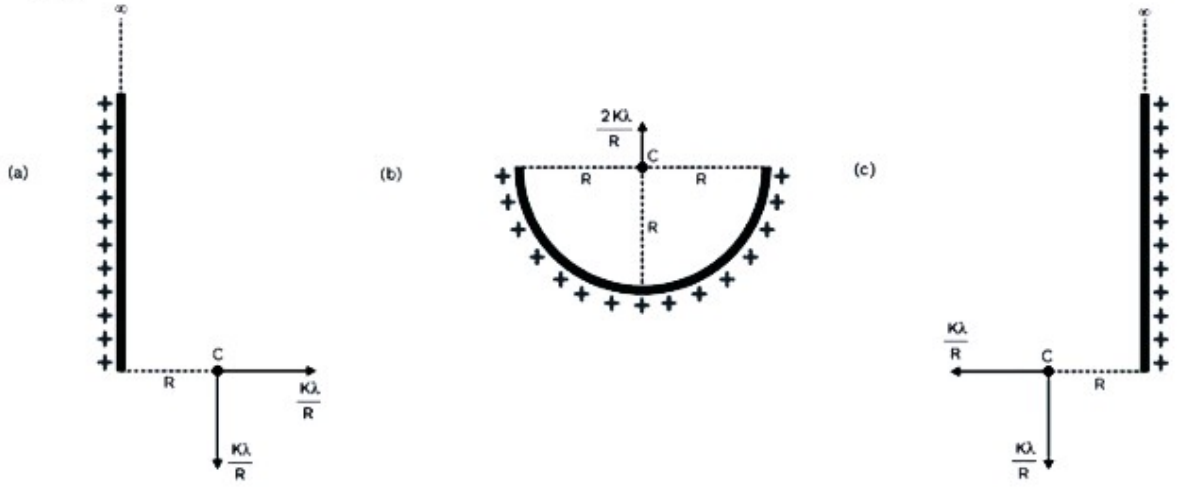


Resultant field at C (a + b + c);

Repeating the same process as in previous question.

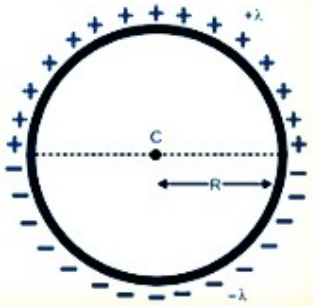


(II)

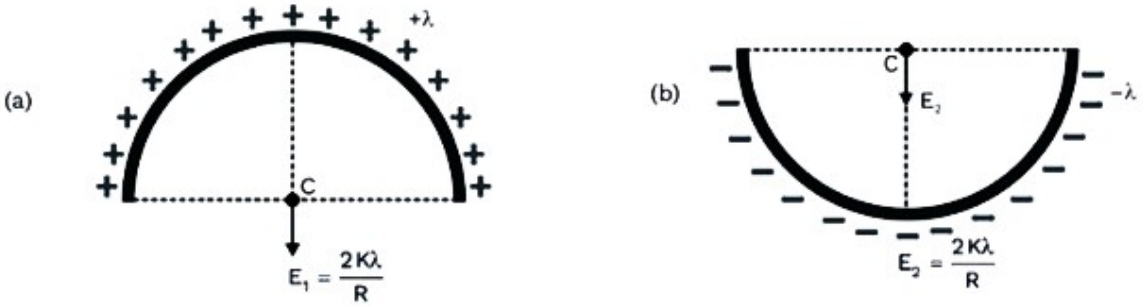


Resultant field at C (a + b + c);
Net electric field at C is zero.

Q. **Find the electric field at centre 'C' in the given case.**



Sol: We can break this ring in two parts and then calculate resultant electric field at C.



Resultant field at C (a + b) will be the vector sum of the electric fields due to the two parts;

$$(E_{net})_C = E_1 + E_2 = \frac{4k\lambda}{R}$$

Electric field due to uniformly charged disc at a point on its axis

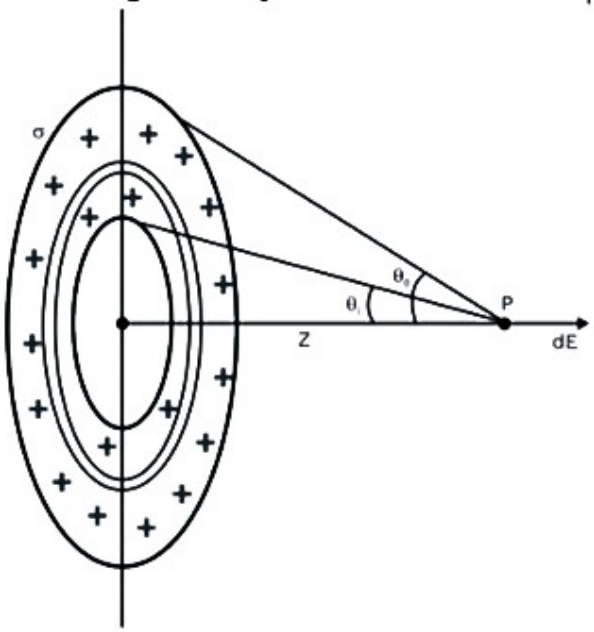
Consider a uniformly charged disc having a surface charge density σ with inner radius r_i and outer radius r_o.

Here, σ = Charge/Area.

θ_i = Angle between inner ring and axis.

θ_o = Angle between outer ring and axis.

Assume an elemental ring of radius r and radial width dr produces an electric field dE at point P,



$$dE = \frac{K(dq)z}{(r^2+z^2)^{3/2}} = \frac{z(\sigma 2\pi r dr)}{4\pi\varepsilon_0(r^2+z^2)^{3/2}}$$

$$\int dE = \frac{\sigma z}{2\varepsilon_0}\int_{r_i}^{r_o}\frac{rdr}{(r^2+z^2)^{3/2}}$$

$$E = \frac{\sigma z}{2\varepsilon_0}\left[\frac{-1}{(r^2+z^2)^{1/2}}\right]_{r_i}^{r_o}$$

$$E = \frac{\sigma}{2\varepsilon_0}\left[\frac{-z}{\sqrt{r_0^2+z^2}} + \frac{z}{\sqrt{r_i^2+z^2}}\right]$$

$$E = \frac{\sigma}{2\varepsilon_0}\left[-\cos\theta_0 + \cos\theta_i\right]$$

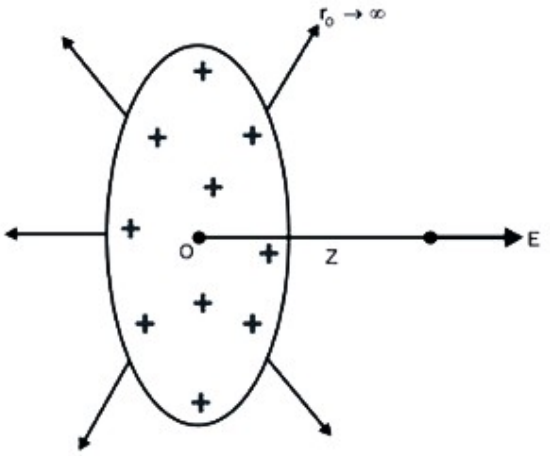
$$E = \frac{\sigma}{2\varepsilon_0}\left[\cos\theta_i - \cos\theta_0\right]$$

Electric field due to large uniformly charged thin layer

For an infinitely large sheet, r_i=0 and $r_o \to \infty$ which means θ_i=0 and $\theta_o \to 90°$.

So, $\cos\theta_i = 1$ and $\cos\theta_o \to 0$.

By putting above values in previous derivation, we get-



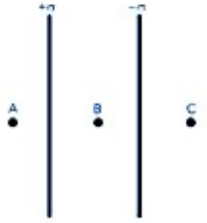
$$E = \frac{\sigma}{2\varepsilon_0}$$

Q. **Find electric field at the given points in the following cases of infinite charged sheets.**

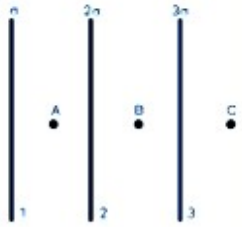
(I)



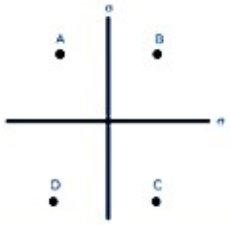
(II)



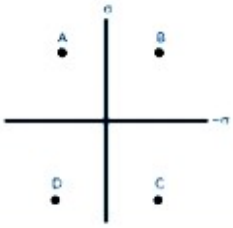
(III)



(IV)

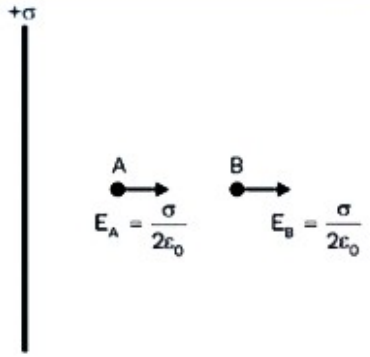


(V)



Sol: (I) $E_A = \dfrac{\sigma}{2\varepsilon_0}$

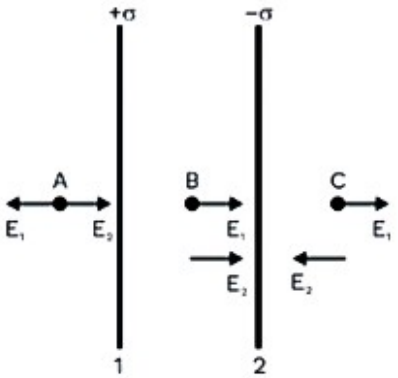
$E_B = \dfrac{\sigma}{2\varepsilon_0}$



(II) $E_A = E_1 - E_2 = \dfrac{\sigma}{2\varepsilon_0} - \dfrac{\sigma}{2\varepsilon_0} = 0$

$E_B = E_1 + E_2 = \dfrac{\sigma}{\varepsilon_0}\ (\rightarrow)$

$E_C = E_2 - E_1 = 0$

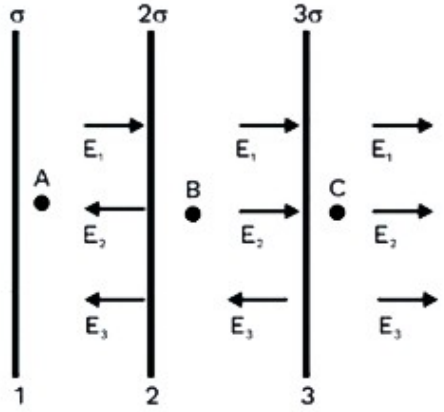


(III) $\vec{E}_A = E_2 + E_3 - E_1$

$$= \frac{2\sigma}{2\varepsilon_0} + \frac{3\sigma}{2\varepsilon_0} - \frac{\sigma}{2\varepsilon_0} = \frac{2\sigma}{\varepsilon_0}(\leftarrow)$$

$$\vec{E}_B = E_1 + E_2 - E_3 = \frac{\sigma}{2\varepsilon_0} + \frac{2\sigma}{2\varepsilon_0} - \frac{3\sigma}{2\varepsilon_0} = 0$$

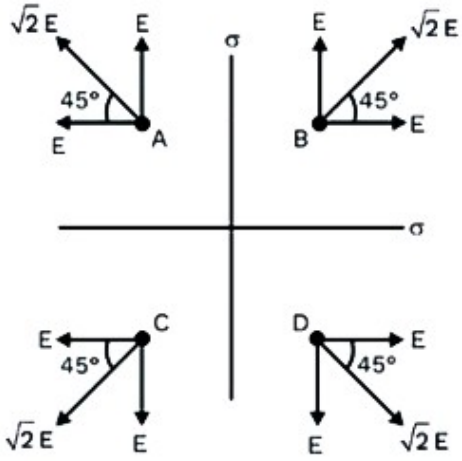
$$\vec{E}_C = E_1 + E_2 + E_3 = \frac{3\sigma}{\varepsilon_0}(\rightarrow)$$



The magnitude of the electric field at every point in part (IV) and part (V) is $\frac{\sqrt{2}\sigma}{2\varepsilon_0}$ with the directions shown in the figures.

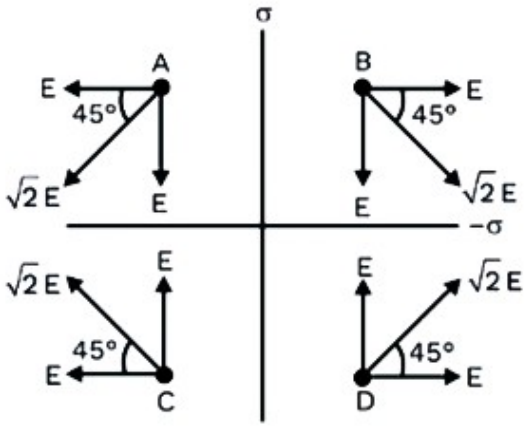
(IV) $E_A = E_B = E_C = E_D = \sqrt{2}E = \frac{\sqrt{2}\,\sigma}{2\,\varepsilon_0}$

= (in the direction as shown in figure)

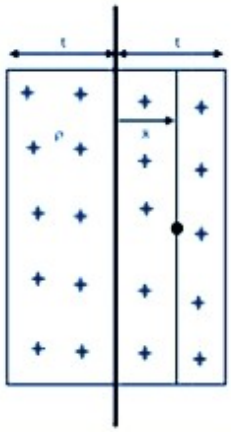


(V) $E_A = E_B = E_C = E_D = \sqrt{2}E = \frac{\sqrt{2}\sigma}{2\varepsilon_0}$

(in the direction as shown in figure)



Q. **Find electric field as a function of x and draw E-x graph for the charged thick sheet as shown. Take positive direction towards right.**



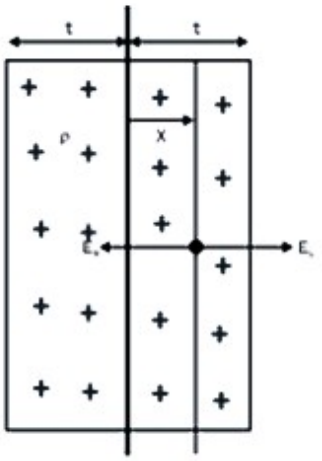
Sol: Field at $x = E_x = E_{left} - E_{Right}$

$$E_x = E_L - E_R = \frac{\sigma_L}{2\varepsilon_0} - \frac{\sigma_R}{2\varepsilon_0}$$

$$E_x = \frac{1}{2\varepsilon_0}\left[\frac{\rho A(t+x)}{A} - \frac{\rho A(t-x)}{A}\right]$$

$$E_x = \frac{1}{2\varepsilon_0}[2\rho x] = \frac{\rho}{\varepsilon_0}x$$

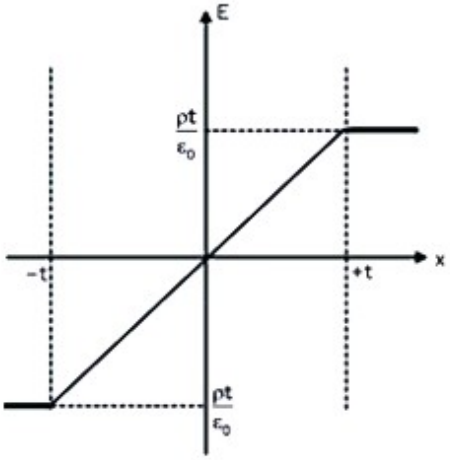
$$E_x = \frac{\rho}{\varepsilon_0}x, \text{ for } (-t \le x \le +t)$$



Outside the sheet,

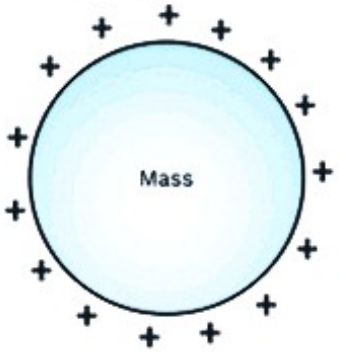
$$E_x = \frac{\rho t}{\varepsilon_0}, \text{ for } (x < -t \text{ or } x > +t)$$

The variation of the electric field can be plotted as,



Electric field due to charged spherical shell

(1) In electrostatics, the word "shell" is always taken in the sense of charge (not in the sense of mass), i.e., shell can have mass inside it but whole charge should be present at the outer spherical surface uniformly.

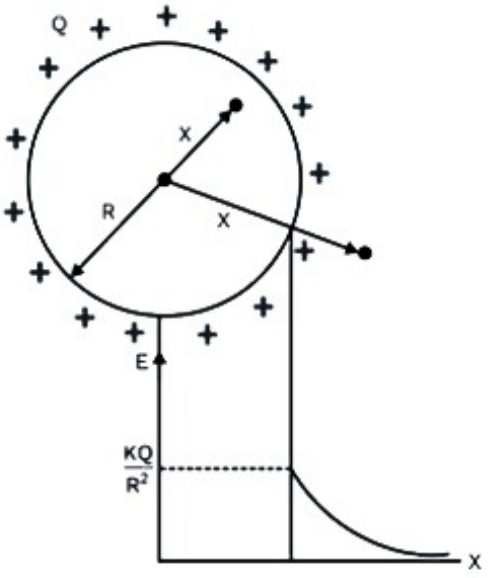


(2) The field inside the uniformly charged spherical shell is always zero due to the charges on the shell. Net electric field inside the shell may or may not be zero.

(3) Uniformly charged spherical shell or solid sphere behaves like a point charge of same magnitude placed at the centre of sphere for outside region.

(4) $E = \begin{cases} 0 & ; x < R \\ \dfrac{KQ}{x^2} & ; x \geq R \end{cases}$

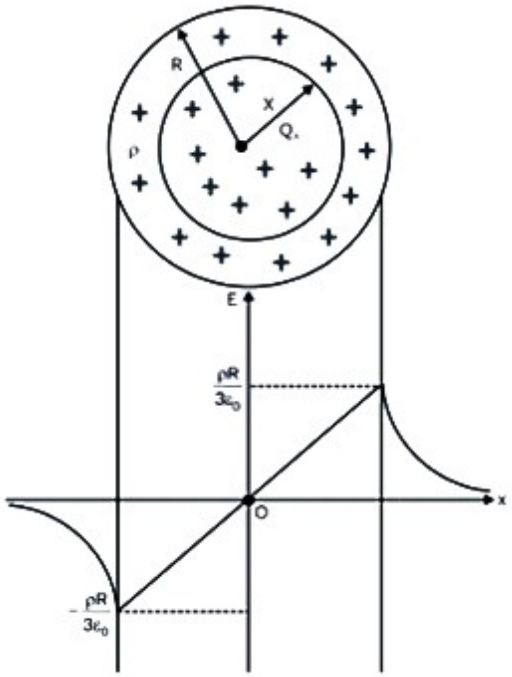
If E is asked on surface then we assume that the point is just outside the surface.



Electric field due to uniformly charged solid sphere

To find the electric field due to a uniformly charged solid sphere at a radial distance x, we have to consider the charge contained within this volume.

If ρ is the volume charge density of the sphere, then



$$Q_r = \frac{4\pi}{3}x^3\rho$$

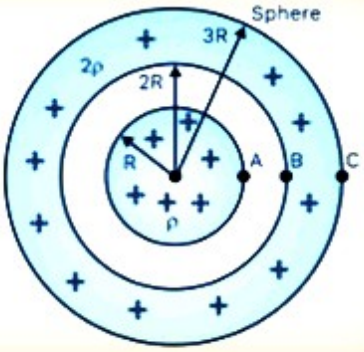
$$\therefore \quad E_x = \frac{KQ_r}{x^2} = \frac{1}{4\pi\varepsilon_0}\frac{4\pi}{3}\frac{x^3\rho}{x^2}$$

$$\vec{E}_x = \frac{\rho}{3\varepsilon_0}\vec{x} \text{ .....(for } x < R).$$

$$E_r = \frac{KQ}{x^2} = \frac{1}{4\pi\varepsilon_0}\frac{\rho\, 4\pi R^3}{3x^2}$$

$$E_r = \frac{\rho R^3}{3\varepsilon_0}\frac{1}{x^2} \text{ ....(for } x \geq R).$$

Q. **Find electric field at the points A, B and C as shown in the given arrangement.**



Sol: For electric field at point A, $E_A = K\frac{Q_{0\to A}}{R^2} = \frac{1}{4\pi\varepsilon_0}\left(\frac{4\pi}{3}R^3\rho\right)\frac{1}{R^2} = \frac{\rho R}{3\varepsilon_0}$

Similarly,

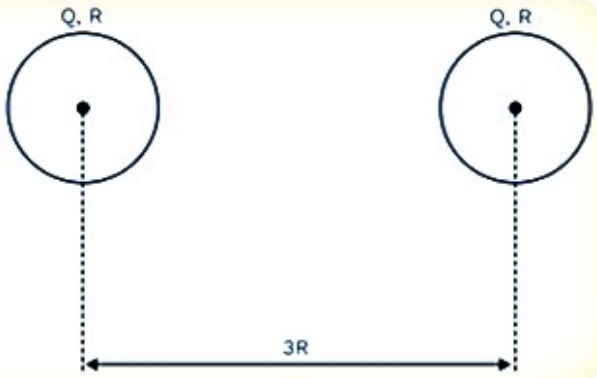
$$E_B = \frac{1}{4\pi\varepsilon_0}\frac{Q_{0\to B}}{(2R)^2} = \frac{1}{4\pi\varepsilon_0}\frac{\left(\frac{4\pi}{3}R^3\rho + 0\right)}{4R^2} = \frac{\rho R}{12\varepsilon_0}$$

, and,

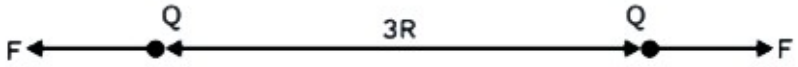
$$E_C = \frac{1}{4\pi\varepsilon_0}\frac{Q_{0\to C}}{(3R)^2} = \frac{1}{4\pi\varepsilon_0(9R^2)}\left[\frac{4\pi}{3}R^3\rho + \frac{4\pi}{3}\left\{(3R)^3 - (2R)^3\right\}2\rho\right]$$

$$E_C = \frac{39\rho R}{27\varepsilon_0}.$$

Q **Find the force acting between two uniformly charged spheres of radius R each the centres of which are at a distance 3R apart.**



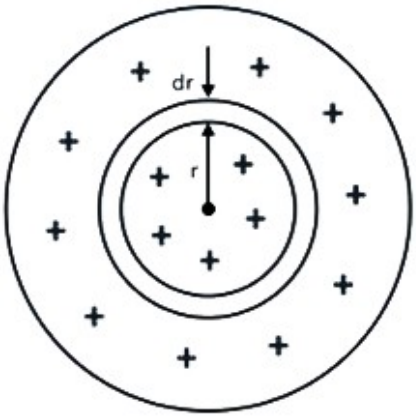
Sol: As we know that electric field due to uniformly charged sphere (outside the sphere) is equal to the field due to same point charge.



So, the force acting between two spheres, $F = \frac{KQ^2}{9R^2}$.

Q. **Find electric field at a radial distance r inside a charged sphere which has a variable volume charge density given by $\rho_r = \rho_0 r$, where ρ_0 is a positive constant.**

Sol: Here, the volume charge density is variable, so we have to apply the integration approach. Consider an elemental shell of radius r and radial thickness dr as shown.



Charge in an elemental shell of radius 'r', thickness 'dr' is

$$dq = \rho_r . 4\pi r^2 dr = 4\pi\rho_0 r^3 dr$$

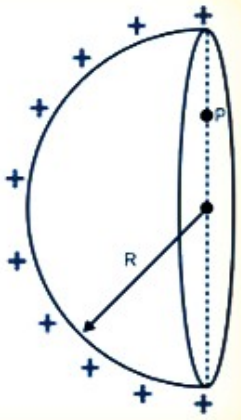
Total charge up to radius r is

$$Q_r = \int dq = 4\pi\rho_0 \int_0^r r^3 dr = 4\pi\rho_0 \frac{r^4}{4}$$

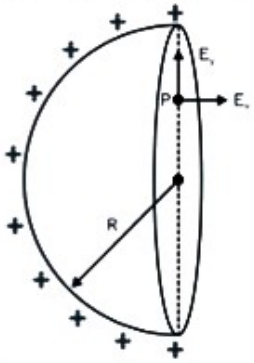
$$\Rightarrow Q_r = \rho_0 \pi r^4$$

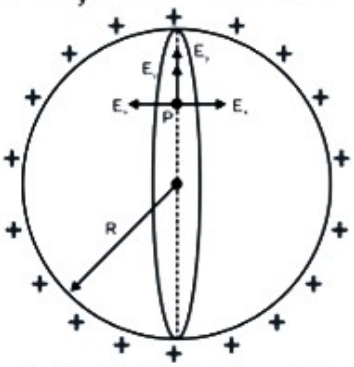
Electric field at r, $E_r = \frac{1}{4\pi\varepsilon_0}\frac{Q_r}{r^2} = \frac{\rho_0 r^2}{4\varepsilon_0}$

Q. **Find direction of electric field at point P in the given uniformly charged hollow hemi-spherical shell.**



Sol: Suppose field has two components E_x and E_y due to hemispherical shell.

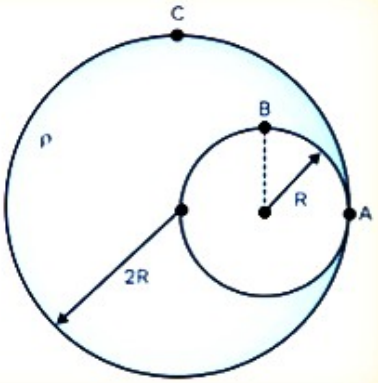




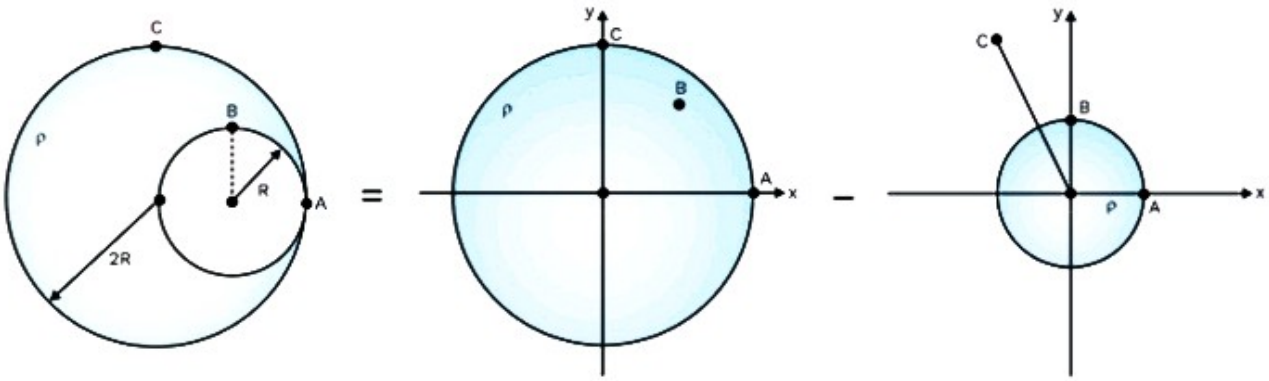
Similarly, the combination of two hemispherical shells will give net field in y-direction.

Since we know that a uniformly charged shell has no net field inside it, therefore $E_y = 0$ and the hemispherical shell will produce electric field along x-direction only.

Q. **Find the electric field at points A, B and C due to the charge in the sphere with cavity as shown in the figure.**



Sol:



$$\vec{E}_A = \frac{\rho\vec{r}}{3\varepsilon_0} = \frac{\rho}{3\varepsilon_0}\left(2R\hat{i}\right) \qquad \vec{E}_A = \frac{\rho}{3\varepsilon_0}\left(R\hat{i}\right)$$

$$\vec{E}_B = \frac{\rho}{3\varepsilon_0}\left(R\hat{i} + R\hat{j}\right) \qquad \vec{E}_B = \frac{\rho}{3\varepsilon_0}\left(R\hat{j}\right)$$

$$\vec{E}_C = \frac{\rho}{3\varepsilon_0}\left(2R\hat{j}\right)$$

$$\vec{E}_C = \frac{1}{4\pi\varepsilon_0}\left(\rho\frac{4\pi R^3}{3}\right)\frac{(-R\hat{i} + 2R\hat{j})}{\left|-R\hat{i} + 2R\hat{j}\right|^3} = \frac{\rho}{15\sqrt{5}\varepsilon_0}\left(-R\hat{i} + 2R\hat{j}\right)$$

Net field at A,

$$\vec{E}_A = \frac{\rho}{3\varepsilon_0}\left(2R\hat{i} - R\hat{i}\right) \Rightarrow \vec{E}_A = \frac{\rho R}{3\varepsilon_0}\hat{i}$$

Net field at B,

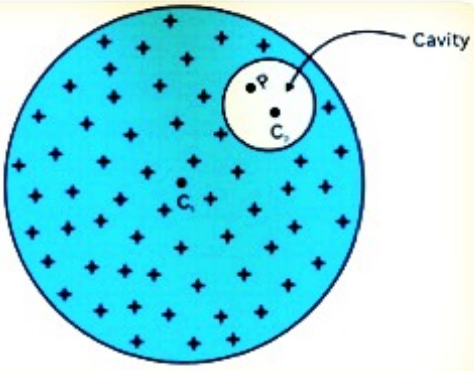
$$\vec{E}_B = \frac{\rho}{3\varepsilon_0}\left(R\hat{i} + R\hat{j} - R\hat{j}\right) \Rightarrow \vec{E}_B = \frac{\rho R\hat{i}}{3\varepsilon_0}$$

Net field at C,

$$\vec{E}_C = \frac{\rho}{3\varepsilon_0}\left(2R\hat{j} + \frac{R\hat{i}}{5\sqrt{5}} - \frac{2R}{5\sqrt{5}}\hat{j}\right)$$

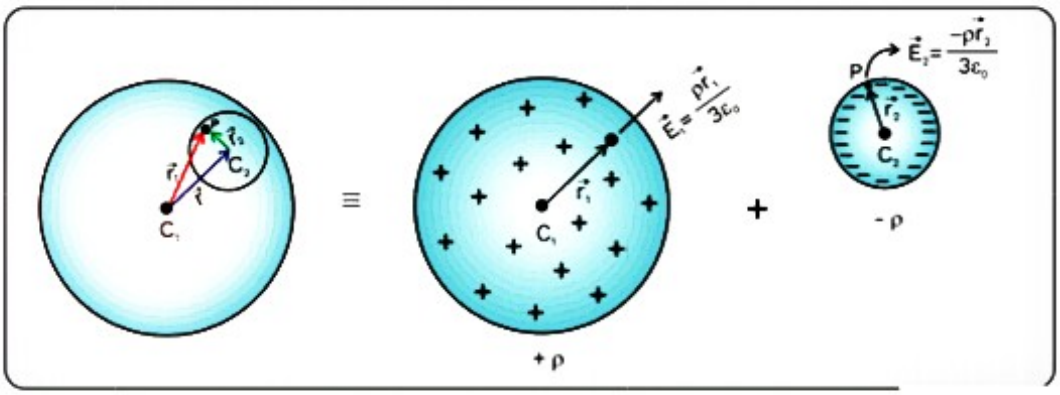
$$\vec{E}_C = \frac{\rho R}{3\varepsilon_0}\left[\frac{1}{5\sqrt{5}}\hat{i} + \left(2 - \frac{2}{5\sqrt{5}}\right)\hat{j}\right]$$

Q. **A spherical portion (cavity) has been removed from the uniformly charged non-conducting sphere as shown in figure.**
The electric field intensity inside the cavity is:
(A) Zero everywhere
(B) Uniform and Constant
(C) Non-uniform
(D) None of these

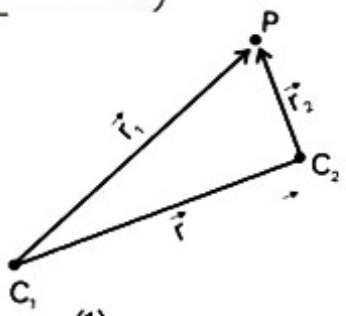


Sol. **(B)**

For solving such type of problems, we use negative charge concept or negative mass concept in case of mass. A charged body with cavity can be replaced by a body with $+\rho$ and a body with $-\rho$.



So, net electric field at point 'P' is given by



$$\vec{E}_p = \vec{E}_1 + \vec{E}_2$$

$$E_p = \left(\frac{\rho \vec{r}_1}{3\epsilon_0}\right) + \left(-\frac{\rho \vec{r}_2}{3\epsilon_0}\right)$$

$$\vec{E}_p = \frac{\rho}{3\epsilon_0}\left(\vec{r}_1 - \vec{r}_2\right) \quad \ldots(1)$$

Use triangle law of vector addition;

$$\vec{r}_1 = \vec{r} + \vec{r}_2 \Rightarrow \vec{r} = \vec{r}_1 - \vec{r}_2 \quad \ldots(2)$$

From equation (1) and equation (2)

$$\vec{E}_p = \frac{\rho}{3\epsilon_0}(\vec{r})$$

$$\boxed{\vec{E}_p = \frac{\rho\left(\overrightarrow{C_1C_2}\right)}{3\epsilon_0} = \text{Constant}}$$

Conclusion:

Inside a cavity of a non-conducting sphere, the electric field is non-zero and constant.

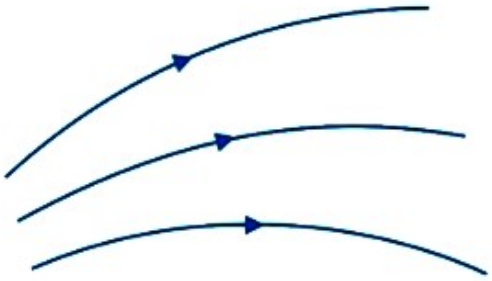
Electric Field Lines

The concept of electric field lines was given by Michael Faraday to visualise electric fields.

Electric field lines are imaginary lines used to represent direction and magnitude of electric field.

Let's understand how electric field lines represent direction and magnitude of electric field.

Suppose electric field lines are drawn as shown in the figure. Then number of field lines per unit area represents magnitude of electric field.

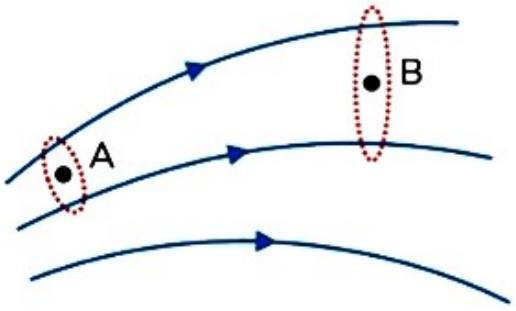


Two field lines are crossing at area taken at A and B. However, area at A is smaller compared to area at B.

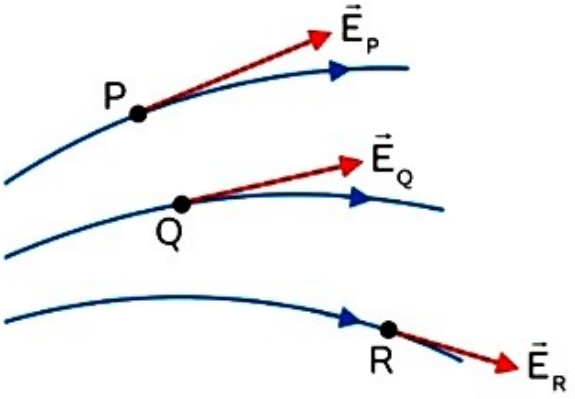
$$\left(\frac{\text{Number of field lines}}{\text{Area}}\right)_A > \left(\frac{\text{Number of field lines}}{\text{Area}}\right)_B$$

$$\Rightarrow \quad |\vec{E}_A| > |\vec{E}_B|$$

$\therefore$ Magnitude of electric field strength at point A is greater than magnitude of electric field strength at point B.

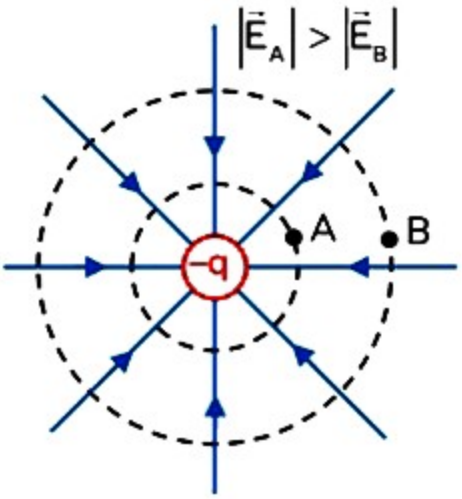


Tangent drawn at any point on electric field lines gives direction of electric field at that point.

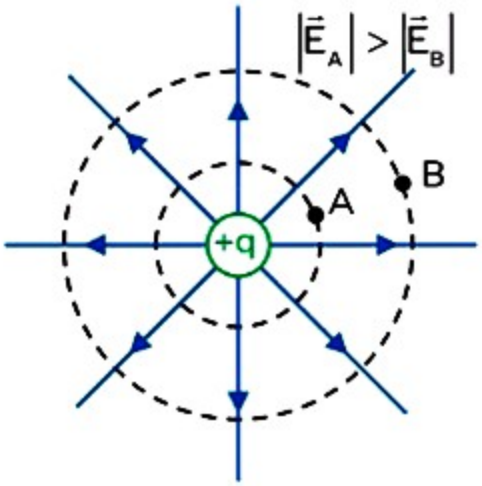


Electric field lines due to isolated point charge

We know that electric field is away from a positive point charge. Electric field lines due to isolated positive point charge are radially outward. As we move away from the charge, same number of lines cross more area. This means density of field lines decreases as we move away from the charge, hence intensity of electric field decreases as we move away from the charge. It is clear from the adjacent diagram that electric field lines originate from a positive charge.



Also, electric field is towards a negative point charge. Here, electric field lines are radially inward. Moving away causes decrease in strength of field. Electric field lines terminate on a negative charge.

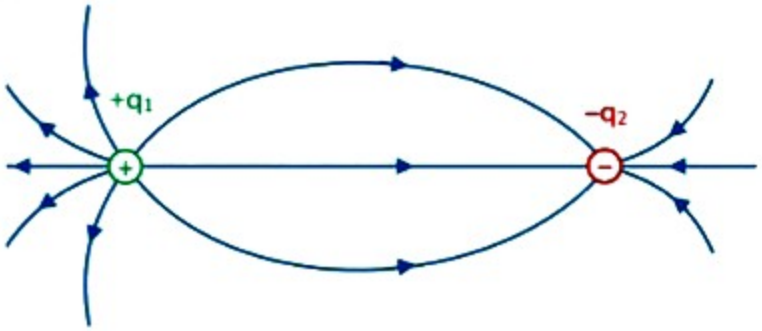


It can be noted that–

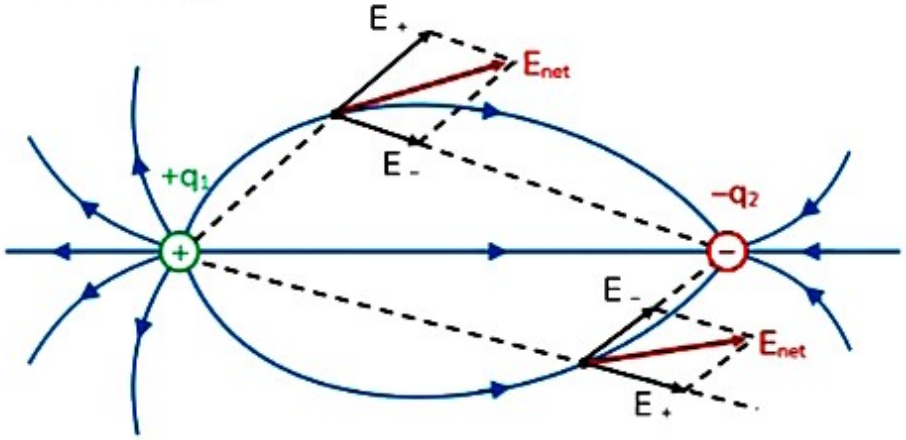
Number of electric field lines originating from or terminating to any charge 'q' is proportional to $\frac{'q'}{\epsilon_0}$.

Electric field lines for two charge (+ve and -ve) system

Electric field lines due to positive and negative point charges placed a distance r apart are shown in the figure below.



We have seen electric field lines of isolated positive or negative charge, and they were straight lines. But here field lines are curved because they represent resultant electric field due to both the charges.



From figure, it can be seen that,

8 lines are originating from charge '+ q_1'.

$\Rightarrow \quad |+q_1| \propto 8 \Rightarrow q_1 \propto 8$

6 lines are terminating on charge '– q_2'.

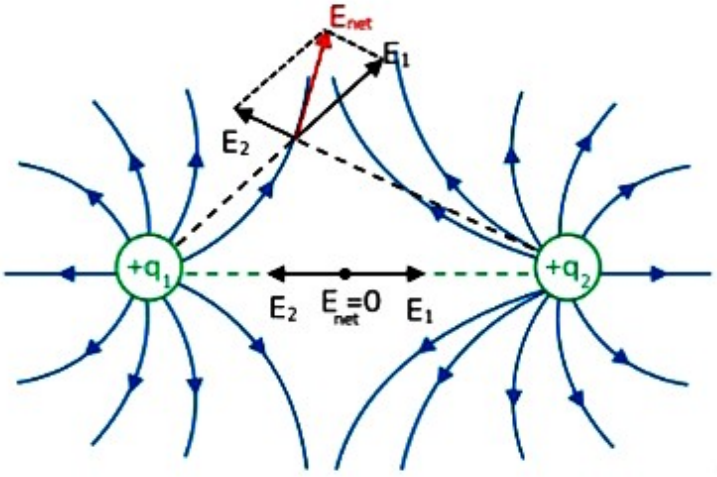
$\Rightarrow \quad |-q_2| \propto 6 \Rightarrow q_2 \propto 6$

$\Rightarrow \quad \frac{q_1}{q_2} = \frac{8}{6} = \frac{4}{3}$

In general, if N_1 lines are associated with charge q_1 and N_2 lines are associated with charge q_2, then

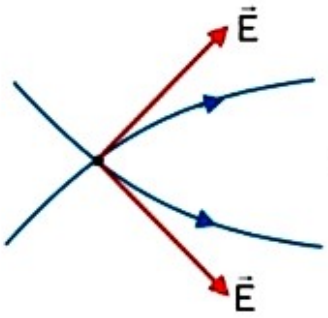
$$\frac{|q_1|}{|q_2|} = \frac{N_1}{N_2}$$

Electric field lines for two positive charges



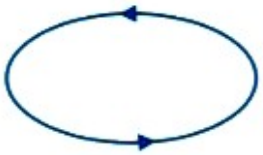
Properties of electric field lines

(i) Electric field lines can never intersect each other. If the field lines intersect, then at the intersection point, electric field will have two directions which is incorrect as electric field can have only one direction at a point.



Impossible

(ii) Electrostatic field lines (electric field lines produced by static charges) never form closed loops.



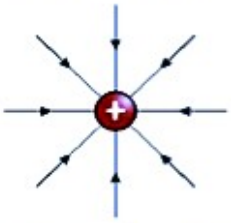
Impossible in electrostatics

However electric field lines due to induced electric field form closed loops. This thing will be discussed in later chapter. Induced electric field is produced by changing magnetic field.

As electric field lines produced by static charges do not form closed loops, electric field due to charge at rest is conservative in nature.

Q. Which of the following diagrams show the correct electric field lines?

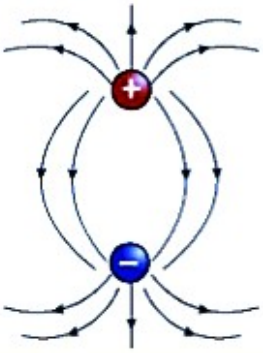
(A)



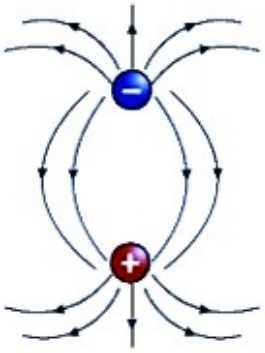
(B)



(C)



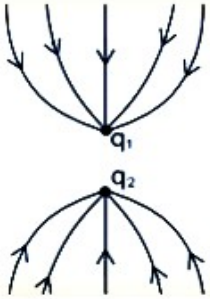
(D)



Sol: **(B)**

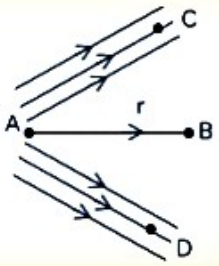
(A) → Field lines are terminating at the +ve charge which is "wrong".
(B) → Field lines are leaving from +ve charge which is "correct".
(C) → Some of the field lines are leaving from –ve charge which is "wrong".
(D) → Some of the field lines are leaving from –ve charge which is "wrong".

Q. **Identify the sign of the charges in the adjoining figure.**



Sol: Both q_1 and q_2 are negative as field lines are terminating on each of them.

Q. **If the electric field at A and B are E_A and E_B respectively, and distance between A and B is r then**
(A) $E_A > E_B$
(B) $E_A < E_B$
(C) $E_A = E_B$
(D) $E_A = E_B/r^2$



Sol: Field lines are more crowded at A than B, so $E_A > E_B$. Therefore option (A) is correct.